Learned representations and kernel corrections for radiative-transfer emulation

Yitzchak Shmalo*

September 2026

Abstract

We compare neural networks, exact Matérn kernel regression and their combinations as emulators of a six-dimensional radiative-transfer table on the 285-band EMIT grid, and we study how their forward accuracy carries over to reflectance retrieval. Over ten partitions of 23,313 states, a kernel with one length scale per input reaches 0.095% mean relative radiance error and a network 0.387%; a kernel on the concatenated hidden features of five networks reaches 0.076%. Forward and inverse rankings disagree. A convex stack improves on a wide feature kernel by only 0.0006 radiance percentage points, yet its 95th-percentile reflectance error is larger at every partition, three times larger on average. We prove that on the physical domain the constrained retrieval error is at most the retrieval-weighted component error divided by the transmission, with constant one, that the rate this implies is sharp, and that errors proportional to the transmission lose no rate. The reversal persists on the physical domain of the retrieval and when every family is retrained on physically admissible targets only; on a second radiative-transfer code, whose transmissions stay away from zero outside one opaque band, it is much weaker.

1 Introduction

Atmospheric correction for imaging spectroscopy relates the radiance measured at the sensor to the reflectance of the surface through radiative-transfer calculations that depend on the state of the atmosphere and on the viewing geometry. Learned emulators replace repeated simulator evaluations with a fitted map from that state to the spectral quantities the retrieval needs. Neural emulators have been built into reflectance retrieval [1], Gaussian-process emulators have been developed for the OCO-2 forward model [3], and physics-guided multi-fidelity networks use a cheaper code as a second source of information [4]. Two questions follow: which representation gives the most accurate forward emulator, and which of its errors matter once the emulator is inverted.

We study exact kernel regression, fully connected networks and two post-hoc combinations of the two. In residual correction the network supplies a mean and a kernel fits its training residual. In feature regression the network supplies a frozen representation and a kernel fits the target on that representation [12]. Both differ from deep kernel learning, where representation and kernel are trained jointly [10], and from regularizing the inner layers of a network with kernel flows [11]. The comparison complements work on kernel methods for operator learning [5], on

*Einstein Institute of Mathematics, The Hebrew University of Jerusalem, Jerusalem, Israel.
.

neural operators such as DeepONet and Fourier neural operators [6, 7], and on classical kernel regression [8].

The data are 23,313 tabulated radiative-transfer evaluations on the 285-band EMIT grid [13]. The state has six coordinates and each output component is well described by a few dozen principal coefficients, so an exact kernel solve on all 18,884 training rows is affordable. On ten common partitions, the kernel with one length scale per input is four times more accurate than the network on radiance, a residual kernel reduces the network's error by a factor of almost three, and an exact kernel on the network's hidden features is about as accurate as the input-scaled kernel. A wider network is a worse emulator but a better feature map: from width 512 to width 2,000 the network's own radiance error doubles while the error of a kernel on its features falls by a fifth. Concatenating the hidden features of five networks and solving one kernel on them improves on the features of a single network at every partition, and among the models fitted to every row of the table on all ten partitions it has the lowest radiance error. Refitting after dropping every training state with an entry outside the physical domain improves the kernel on the network's features on every partition at both widths, and at width 512 it does so while the network itself gets worse, as widening does (Section 6.3).

The central empirical result concerns the inverse. The stack built on the wide network improves mean radiance error over its own feature kernel by 0.0006 percentage points, while its mean within-split 95th-percentile reflectance error rises from 7.57 to 22.71 points, and the tail is worse at all ten partitions. Small forward gains can therefore coexist with large changes in the tail of the retrieval error. The reversal is not produced by the few entries where the inverse is undefined: it persists on the physical domain of the retrieval, with coverage and failure counts reported, after retraining on physically admissible targets only, and on a fresh partition evaluated once under a hypothesis fixed in advance. On a second radiative-transfer code, whose transmissions stay above 0.14 away from one opaque band, it is much weaker (Section 6.6).

The analysis has three levels. Exact finite-perturbation identities relate radiance and reflectance errors through the inverse denominator. A transfer inequality with constant one bounds a physically constrained retrieval by the component error relative to the transmission. Tail and conditional-error assumptions then give rates, including a sharp worst-case exponent and conditions under which no rate is lost; the central quantity is the profile of the conditional error against the transmission, not the error alone. On test data the transmission-weighted term of the bound ranks nine emulators exactly as their 95th-percentile retrieval errors, which the unweighted component error does not. For residual correction, a Hilbert-space identity gives the exact criterion for improvement, and a principal-component decomposition separates truncation error from the error of the kernel step.

Section 2 reviews related work and Section 3 describes the data and the protocol. Sections 4 and 5 give the models and the main comparison, and Section 6 the physical domain, the training targets, a fresh partition and a second radiative-transfer code. Section 7 varies training size, output rank and width. Sections 8 and 9 develop the residual and retrieval analyses, and Sections 10 and 11 test them on the data. Proofs, further experiments, a comparison on OCO-2 and ten further emulation problems are in the supplement, whose sections, results, tables and equations carry the prefix S.

2 Related work

Radiative transfer and its surrogates. Relevant radiative-transfer codes include 6S [14], libRadtran [15, 16] and MODTRAN [17]; lookup-table calculations enter atmospheric correction [2] and optimal estimation [18]. The carbon-dioxide product of OCO-2 [19, 20] rests on the forward model and inverse method of [21, 22]. Surrogates enter because the forward model is the cost. Neural emulation of longwave radiation goes back to [23] and was carried into climate models by [24]; later work measures how accurate such emulators can be made [25, 26] and how learned parameterizations behave when the host model is stepped forward [27, 28]. In imaging spectroscopy and land-surface inference the same idea appears as emulator toolboxes for radiative-transfer models [29, 30], Gaussian-process emulation for parameter retrieval [32], and the wider use of Gaussian processes in earth observation [33]. For the transfer functions of atmospheric correction, [31] compared regression algorithms on MODTRAN simulations, found Gaussian processes and kernel ridge regression the most accurate, and used the best emulator for the correction itself, with its largest reflectance errors in the absorption bands; [1] places an emulator inside a reflectance retrieval and [3] fits a cross-validated Gaussian process to the OCO-2 forward model. Replacing an expensive simulator by a statistical model is a classical problem [34], as are calibrating it against observations [36] and using a cheap model to inform an expensive one [35, 37], surveyed for propagation and inference in [38]; the network of [4] is a multi-fidelity instance on the corpus rescored in Section S4. The measurement requirements behind the tables used here are those of the EMIT mission [13] and of the surface-imaging missions that follow it [39].

Kernel regression. The estimators are classical: interpolation in a reproducing-kernel Hilbert space [40], smoothing splines and the representer theorem [41, 42, 43], ridge regularization [44] read as spectral filtering of an ill-posed problem [45], and the Matérn family, whose smoothness index fixes the differentiability of the sample paths [46, 8]. Rates under source and capacity conditions are given in [47, 48, 49, 50]; the textbook accounts are [51, 52, 53]. Each component here is a vector of principal coefficients, so the multi-output formulation is the relevant one [54, 55]. Kernels also serve as solvers rather than regressors: Bayesian numerical homogenization [56], operator-adapted wavelets [57] and the Gaussian-process treatment of nonlinear partial differential equations [58] use the same estimator, and cross-validating the kernel itself gives the kernel-flow family [59, 60, 11].

Cost. An exact solve is cubic in the number of training rows, which has produced a long line of approximations: the Nyström method and its analysis [61, 62, 63], random features [64] and their operator-learning counterpart [65], preconditioned solvers at large scale [67, 68], inducing-point and variational Gaussian processes [69, 70, 71], and structured interpolation with hardware-aware linear algebra [72, 73]. At $n = 18{,}884$ the exact factorization is affordable, and every full-data EMIT fit below is an exact solve.

Choosing and combining kernels. A kernel can be built rather than picked: spectral mixtures [74], additive structure [75], compositional search over a grammar [76], and per-coordinate relevance fitted by marginal likelihood [8] or by cross-validation [59]. Several kernels can be combined instead of one being chosen, by semidefinite programming [77], by the algorithms surveyed in [78], or by alignment with the target Gram matrix [79, 80]. Combining predictors

rather than kernels is older still [81, 82, 83]. Selection on held-out rows is classical [84, 85, 86]; Section S1.2 records the finite-class guarantee it provides when the selection sample is independent of every earlier fitting decision.

Networks, and the kernels they induce. Feedforward networks are universal approximators [87]; those trained here use the standard adaptive optimizer [88] with decoupled weight decay [89] and early stopping [90]; dropout [91] is another standard regularizer. In the infinite-width limit a network is a Gaussian process [92, 93, 94] whose kernel is the arc-cosine kernel of [95], and gradient descent on a wide network follows the neural tangent kernel [96, 97]; both enter the kernel comparison of Section S2 as heads. The limit is not the whole story, since a finite network learns its representation [99], and the comparison between kernel machines and trained networks is itself unsettled [98]; a learned representation under a kernel is what deep kernel learning [10], the Gaussian process with a neural-operator mean [9] and the construction of [12] exploit. Kolmogorov–Arnold networks [100, 101] are the architecture behind the multi-fidelity model of [4]. Operator learning supplies the other comparison: DeepONet [6], Fourier neural operators [7] and the general formulation [102], with fair-comparison protocols [103], physics-informed training [104], the cost-accuracy trade-off [105], kernel competitors [5], and the benchmark collections [106, 107, 108] from which the supplement (Section S7) draws.

3 Data and protocol

The data are $n = 23{,}313$ tabulated radiative-transfer evaluations provided by the Jet Propulsion Laboratory as a lookup table for EMIT atmospheric correction. Each input $x \in \mathbb{R}^6$ collects the atmospheric and viewing state: an aerosol optical depth in $[0.0500, 1.2000]$, a surface elevation in $[-0.4997, 5.9998]$ km, a water-vapour column in $[0.0502, 6.9999]$ cm, a relative azimuth in $[0.0029, 179.9947]^\circ$, a solar zenith angle in $[0.0012, 64.9981]^\circ$, and a view zenith angle in $[150.0003, 179.9991]^\circ$ from the downward vertical, that is, within 30° of nadir; these are the sampled extrema, over the ranges on which such tables are built [2, 17]. Each output consists of four spectral components on the 285-point EMIT grid spanning 381–2493 nm: path radiance Y_1, direct flux Y_2, diffuse flux Y_3 and spherical albedo Y_4 (Figure 1). The quantities used downstream are the radiance at a surface reflectance ρ,

$$L(\rho) \;=\; Y_1 + \rho\,\frac{Y_2 + Y_3}{1 - \rho\, Y_4}, \tag{1}$$

and the reflectance recovered by inverting (1) with predicted components at the true radiance,

$$\hat{\rho} \;=\; \frac{L - \hat{Y}_1}{\hat{Y}_2 + \hat{Y}_3 + \hat{Y}_4 (L - \hat{Y}_1)}, \tag{2}$$

both evaluated at $\rho = 0.7$ in the main comparison. The table carries the states, the four components and the wavelength grid; the simulator version and its configuration are not part of it, and nothing below depends on them.

Protocol. The table is split into a test block of 2,331 rows and a remaining block of 20,982, of which 2,098 rows are held out for validation and 18,884 are used for fitting; the split is repeated with seeds 101–110. Standardization and principal components are fitted on the training rows

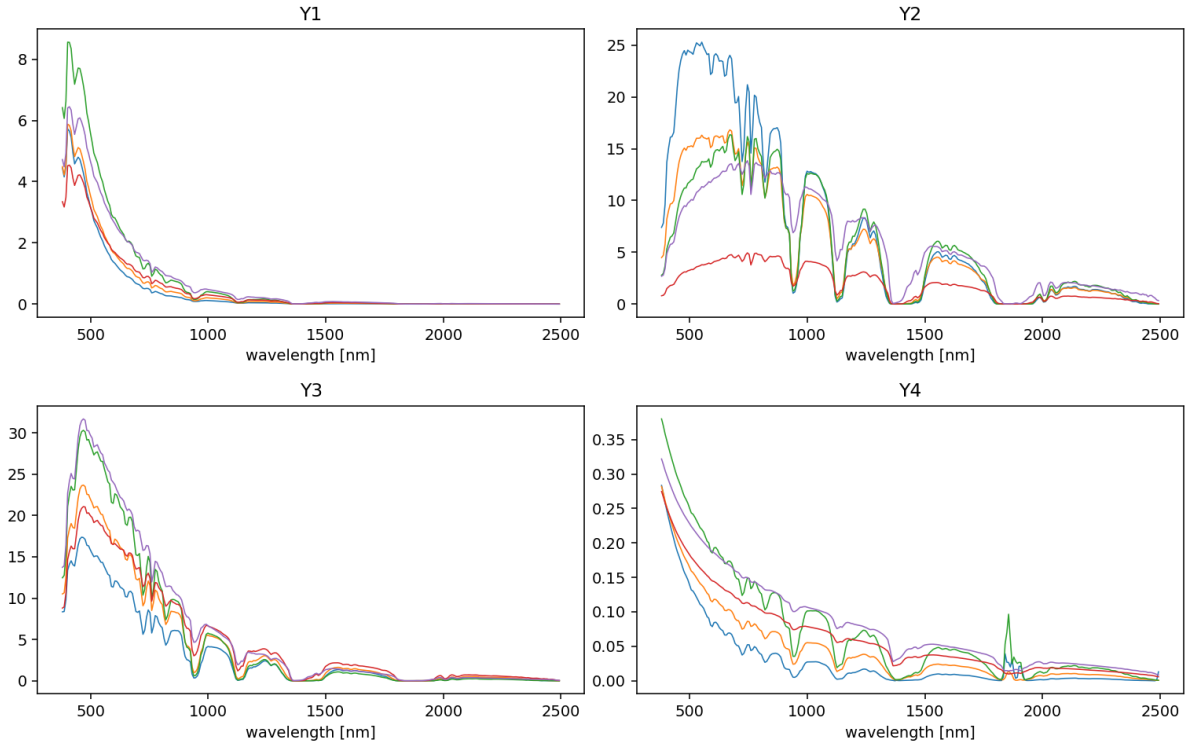


Figure 1: Example output curves across the EMIT wavelength grid, five samples per panel. The curves are smooth, share the deep water-vapor absorption features near 1380 and 1900 nm, and differ mainly by smooth rescalings along the state axes.

of each split, hyperparameters, checkpoints, coordinatewise selections and combination weights are chosen on its validation rows, and every family is evaluated on its test rows. Unless stated otherwise, tables report the mean and sample standard deviation over the ten splits.

The model families and ablations were developed on this table with its test errors in view. The ten splits therefore measure how the comparison varies with the partition; their test blocks overlap, and their standard deviations are descriptive rather than confidence intervals. Section 5.1 reports paired differences on common splits. Three evaluations in Section 6 were specified before they were run: an experiment on the training targets, whose arms, seeds and scoring were recorded in a protocol before its first fit; a fresh partition with a single hypothesis fixed in advance and evaluated once; and the same pipeline on a second radiative-transfer code, with three hypotheses and their decision rule fixed before its first fit.

Structure of the outputs. Three properties of the table shape the comparison. First, the outputs are low rank. On standardized training outputs, 99.99% of the variance of every component lies in the first 11–25 principal components and 99.9999% in at most 44 (Figure 2, left). Neighbouring bands are nearly collinear: the median correlation between adjacent bands is above 0.998 for every component, and the smallest, 0.85 to 0.89, falls at the edges of the water-vapour absorption, between 1350 and 1357 nm for the path radiance and the fluxes and between 1863 and 1871 nm for the albedo, inside the regions that carry the retrieval walls of Section 6. All regressors below act on the first 64 principal coefficients, whose unexplained

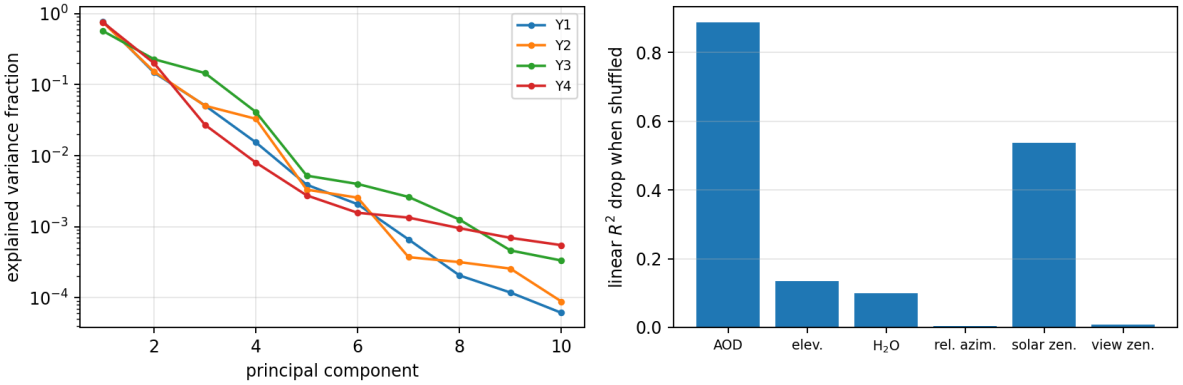


Figure 2: Left: explained-variance spectra of the standardized outputs. Every component concentrates 99.99% of its variance in 11–25 principal components out of 285 bands. Right: drop in linear validation R^2 when a single standardized input coordinate is shuffled. Aerosol optical depth and solar zenith angle dominate; the azimuth and view-zenith coordinates are almost inert at first order.

training variance is below 10^{-7}, except the networks, which are trained on the 285 standardized bands. A small variance tail is an aggregate statement; Section 8 states the projection error explicitly.

Second, the inputs matter unequally. Shuffling the aerosol coordinate reduces the validation R^2 of a linear model by 0.89, the solar zenith angle by 0.54, and elevation and water vapor by about 0.1 each; the two remaining angles have smaller effects (Figure 2, right). This motivates a separate length scale for each input.

Third, low-degree surrogates are already accurate. Linear regression on the standardized state reaches $R^2 = 0.83$, against 0.74 for distance-weighted five-nearest-neighbors, and cubic ridge regression has a mean relative component error of about two percent (Table 1).

Conditioning of the inverse. In deep absorption bands the transmitted flux $Y_2 + Y_3$ can be as small as 10^{-21}, and at a fully opaque band the radiance does not depend on the surface reflectance at all, so the inverse is not identifiable there. Even the exact tabulated components do not invert exactly in floating point: over the whole table the round trip has reflectance RMSE 7.4×10^{-3}, all of it from the entries with the smallest transmissions (Section 6). We therefore report both the center and the tail of the reflectance error. In the all-band statistics a non-finite inverse is assigned zero and counted; none occurs in the main comparison.

Differentiating (2) separates the components. Write $D = \hat{Y}_2 + \hat{Y}_3 + \hat{Y}_4(L - \hat{Y}_1)$ for its denominator. Then

$$\frac{\partial \hat{\rho}}{\partial \hat{Y}_4} = -\hat{\rho}^{\,2}, \qquad \frac{\partial \hat{\rho}}{\partial \hat{Y}_2} = \frac{\partial \hat{\rho}}{\partial \hat{Y}_3} = -\frac{\hat{\rho}}{D}, \qquad \frac{\partial \hat{\rho}}{\partial \hat{Y}_1} = -\frac{\hat{Y}_2 + \hat{Y}_3}{D^2}. \tag{3}$$

At exact identifiable components $\hat{\rho} = \rho = 0.7$, so the albedo derivative is -0.49 whatever the transmitted flux, while the path-radiance and flux derivatives grow without bound as D decreases. Errors in path radiance and flux are therefore the ones that matter near opaque bands.

3.1 Finite-perturbation control of the inversion

The derivatives in (3) describe local conditioning. The following elementary identities give a finite-error test and distinguish forward accuracy from inversion stability. For one sample and band, write $a = Y_1$, $t = Y_2 + Y_3$, $s = Y_4$ and $L = a + \rho t/(1 - \rho s)$. Set $e_a = \widehat{a} - a$, $e_t = \widehat{t} - t$, $e_s = \widehat{s} - s$.

Proposition 1 (Conditional radiance and reflectance stability)**.** *Suppose* $0 \le \rho \le \rho_{\max}$, $t > 0$ *and* $q = 1 - \rho s > 0$. *If also* $\widehat{q} = 1 - \rho\widehat{s} > 0$, *the forward error satisfies the exact identity*

$$\widehat{L} - L = e_a + \frac{\rho e_t}{\widehat{q}} + \frac{\rho^2 t e_s}{q\widehat{q}}. \tag{4}$$

In particular, if $q, \widehat{q} \ge \kappa > 0$,

$$|\widehat{L} - L| \le |e_a| + \rho_{\max}|e_t|/\kappa + \rho_{\max}^2|te_s|/\kappa^2.$$

For inversion at the true radiance, put $u = L - a = \rho t/q$, $D = t/q$, *and* $\widehat{D} = \widehat{t} + \widehat{s}(L - \widehat{a})$. *Whenever* $\widehat{D} \ne 0$,

$$\widehat{\rho} - \rho = -\frac{(1 - \rho\widehat{s})e_a + \rho e_t + \rho u e_s}{\widehat{D}}, \qquad \widehat{D} - D = e_t + u e_s - \widehat{s} e_a. \tag{5}$$

If $B = |e_t| + |ue_s| + |\widehat{s}e_a| \le \eta D$ *for some* $0 \le \eta < 1$, *then* $\widehat{D} \ge (1 - \eta)D > 0$ *and*

$$|\widehat{\rho} - \rho| \le \frac{|1 - \rho\widehat{s}|\,|e_a| + \rho|e_t| + \rho|ue_s|}{(1 - \eta)D}. \tag{6}$$

Proof in Section S9. When $\widehat{q} \ne 0$, the same calculation gives the exact forward–inverse link

$$\widehat{\rho} - \rho = -\frac{\widehat{q}}{\widehat{D}}(\widehat{L} - L). \tag{7}$$

Consequently, on a common set where $|\widehat{q}| \le Q$ and $|\widehat{D}| \ge \delta > 0$, the corresponding unweighted L^2 errors obey $\|\widehat{\rho} - \rho\|_2 \le (Q/\delta)\|\widehat{L} - L\|_2$. Squaring the link gives $(\widehat{L} - L)^2 = (\widehat{D}/\widehat{q})^2(\widehat{\rho} - \rho)^2$ at every entry, so a mean squared radiance error is a mean squared reflectance error in which each entry carries the weight $(\widehat{D}/\widehat{q})^2$, and this weight tends to t^2/q^4 as the component errors vanish. Relative to the retrieval, the forward metric discounts an entry by the square of its transmission, and two emulators whose reflectance errors sit at different transmissions can be ordered one way by radiance and the other way by reflectance. Without the denominator condition there is no uniform implication from small spectral relative radiance error to small reflectance error. For example, take two bands with $a = s = 0$, $t = (1, \varepsilon)$ and $0 < \rho < 1$. A predictor with $\widehat{t} = t$, $\widehat{s} = 0$ and $\widehat{a} = (0, c\varepsilon)$, where $0 < c < \rho$, has relative spectral radiance error $c\varepsilon/(\rho\sqrt{1 + \varepsilon^2}) \to 0$ but retrieves $(\rho, \rho - c)$, so its reflectance RMSE remains $c/\sqrt{2}$. Every denominator is positive for $\varepsilon > 0$ and both retrieved reflectances remain physical. This example isolates the loss of inversion stability as the second-band flux vanishes.

The forward bound need not deteriorate when t tends to zero, whereas the inverse bound requires error control relative to $D = t/q$. Thus a small spectral radiance error does not certify stable reflectance recovery. At $t = 0$, $L = a$ is independent of ρ wherever the forward formula is defined, so reflectance is unidentifiable from this band. The identities are exact-arithmetic statements; floating-point cancellation in $L - a$ can further affect an implemented inverse.

An empirical comparison should therefore fix its passband and conditioning threshold before the models are evaluated. The scorer used in Section 6 takes a positive flux scale from the training rows and a mask computed from the true components alone, common to all families, so that a failing model cannot change its own coverage; it reports the coverage, the non-finite inversions and the denominator failures separately from the error on the successful inversions.

4 Models

All models are trained per component and evaluated in physical units after undoing the standardization. Unless stated otherwise they regress the 64 leading principal coefficients of the standardized component. Every hyperparameter named below is selected on the validation rows of the split at hand. The comparisons of Sections 5–7 and of Section S5 run on CPUs; most of the diagnostics of Sections 10 and 11 and of Sections S2–S4 train their networks on one T4 GPU.

Reference family. Cubic polynomial ridge regression [44] on all monomials of degree at most three in the six standardized inputs, with the ridge weight selected from $\{10^{-10}, 10^{-8}, 10^{-6}, 10^{-4}, 10^{-2}\}$.

Exact kernel ridge regression (KRR). Three constructions, all Matérn kernels [46] on the standardized inputs, with the length scale expressed as a multiple of the median pairwise distance of the fit set. The nugget γ enters the system as $(K+n\gamma I)\alpha = Z$. The subset baseline is a Matérn-5/2 kernel fitted on a uniformly drawn subset of 4,000 training points, the landmark choice of a Nyström approximation [61, 62], with multiplier in $\{1, 2\}$ and nugget in $\{10^{-8}, 10^{-6}, 10^{-4}, 10^{-2}\}$. The second is the exact solve on every training row: smoothness $\nu \in \{3/2, 5/2\}$, multiplier in $\{0.5, 1, 2, 4\}$ and the same nugget grid, tuned on validation from a 6,000-row training subsample, with the winner refit on all 18,884 rows by one Cholesky factorization [109]. The third gives each input its own length scale, an automatic relevance determination [92, 8]: starting from the isotropic winner, a coordinate search over per-input multipliers in $\{1/4, 1/2, 1, 2, 4\}$ runs for two sweeps scored on validation, after which smoothness, scale and nugget are re-tuned at the metric found and the solve is refit on every row.

Fully connected networks (FC-DNN). Three ELU hidden layers of 512 units per component, trained on the 285 standardized bands with AdamW [88, 89] at learning rate 10^{-3}, cosine decay, batch size 1024 and weight decay 10^{-6}, for up to 150 epochs with early stopping on validation loss [90] (patience 25). Five networks are trained per component and split from different initializations; the tables report the first and the mean of the five.

Kernel corrections of the neural mean. The construction of [12], with every kernel stage an exact solve on all training rows. (i) *Residual correction*: an exact Matérn regression, on the standardized inputs, of the network's training residuals in coefficient space, added back to the network's projection into that space, with smoothness, scale and nugget selected by the validation error of the corrected prediction; it is applied to the single network and to the five-network mean. (ii) *Kernel on features*: the same exact regression with the input metric replaced by the network's last hidden layer, standardized. This is the post-hoc counterpart of deep kernel learning [10], in which representation and kernel are trained together, and of the kernel-flow regularization of a network's inner layers [11]; here the representation is fixed first and the kernel is solved exactly

	component rel. L^2 [%]				radiance	reflectance
model	Y_1	Y_2	Y_3	Y_4	rel. L^2 [%]	med. $\lvert\hat{\rho} - \rho\rvert$ [pp]
Cubic ridge	1.16±0.02	2.18±0.04	1.11±0.03	2.98±0.06	0.760±0.011	0.471±0.010
Matérn KRR, 4000-point fit	0.17±0.01	0.61±0.04	0.29±0.02	2.44±0.87	0.332±0.029	0.178±0.030
Matérn KRR, exact on all rows	0.08±0.01	0.42±0.03	0.20±0.00	2.62±0.18	0.239±0.016	0.141±0.018
Matérn KRR, per-input scales	0.08±0.01	0.16±0.02	0.08±0.01	2.18±0.16	0.095±0.009	0.036±0.004
FC-DNN (3×512)	0.48±0.02	0.97±0.04	0.41±0.02	3.02±0.18	0.387±0.012	0.245±0.008
Five-network mean	0.37±0.01	0.67±0.03	0.28±0.01	2.51±0.09	0.251±0.006	0.163±0.003
DNN + residual KRR	0.10±0.00	0.30±0.02	0.16±0.01	2.43±0.08	0.141±0.006	0.089±0.008
Ensemble + residual KRR	0.08±0.00	0.27±0.03	0.13±0.00	2.42±0.17	0.123±0.007	0.080±0.008
Kernel on network features	0.18±0.01	0.17±0.01	0.10±0.01	1.99±0.08	0.098±0.007	0.054±0.004
Per-coordinate selection	0.19±0.01	0.44±0.10	0.13±0.02	2.25±0.18	0.193±0.022	0.123±0.023
Convex stack	0.07±0.00	0.16±0.01	0.09±0.00	1.94±0.06	0.087±0.006	0.048±0.003

Table 1: Test errors under the protocol of Section 3, mean ± sample standard deviation over ten splits. Component and radiance columns report mean relative L^2 error; radiance is evaluated at $\rho = 0.7$. The last column is the median absolute reflectance error in percentage points, over all bands, deep absorption windows included. Table 3 gives the corresponding 95th percentiles and RMSE.

on it. (iii) *Per-coordinate selection*: for each of the 64 output coefficients, the validation winner among the exact kernel, the network, the ensemble, the two corrected predictors and the feature kernel. (iv) *Convex stack*: per component, a convex combination of the same six heads with weights fitted on the validation rows in the physical relative metric, in the manner of stacked generalization [82] and stacked regression [83].

Sample size, rank and width. Section 7 repeats the isotropic exact kernel, the network, its residual correction and the feature kernel at nested training sizes, at other output ranks and at other widths; the main table is the full-size, rank-64, width-512 cell of that grid.

5 Results

Table 1 compares the model families at 18,884 training rows, output rank 64 and network width 512.

Polynomial and subset-kernel baselines. Cubic ridge regression gives 1.86% mean component error and 0.760% radiance error. The 4,000-point Matérn fit reduces these to 0.88% and 0.332%. Its albedo error depends strongly on the split: about 1.9% on seven splits and 3.3–4.3% on the other three.

Full-data and input-scaled kernels. The isotropic kernel on all training rows reduces the radiance error to 0.239% and the path-radiance error to 0.08%, against 0.17% for the subset fit. Its albedo error is 2.3–2.9%: steadier than the subset fit, but larger than that fit on its seven best splits. A separate length scale for each input brings the radiance error to 0.095%, the mean component error to 0.63% and the median reflectance error to 0.036 percentage points. The fitted scales reflect the weak azimuth dependence of the flux components: the relative-azimuth length scale is sixteen to thirty-two times the median of the other five, with median factors of sixteen for direct flux and albedo and thirty-two for diffuse flux, on every split for the flux components and

on nine of ten for albedo. In each of these twenty-nine cells the search has lengthened that scale sixteenfold, the most its two sweeps allow, so the true factors may be larger. The view-zenith scales are four to sixteen times longer for the fluxes and the albedo, and the path-radiance metric is nearly isotropic.

Networks and kernel corrections. The network has 1.22% mean component error and 0.387% radiance error; averaging five initializations gives 0.96% and 0.251%. Residual kernel regression reduces the radiance error of the single network to 0.141% and that of the ensemble to 0.123%. The feature kernel has 0.61% mean component error and 0.098% radiance error, level with the input-scaled kernel: it is lower on four of ten splits, and its mean is slightly higher. Its median reflectance error, 0.054 points against 0.036, is larger on every split.

Selection and stacking. Both combiners use the isotropic full-data kernel, the network, its five-member ensemble, the two residual-corrected predictors and the feature kernel; the input-scaled kernel is not in the pool. Coordinatewise selection gives 0.193% radiance error, worse than several of its members. The convex stack gives 0.087% radiance error, 0.57% mean component error and 0.048 points of median reflectance error, about 8% below the input-scaled kernel and 11% below its best member, the feature kernel, on radiance. On mean component error it improves on the input-scaled kernel at nine splits, by 0.06 ± 0.05 points, and on the feature kernel at all ten, by 0.047 ± 0.006 points.

The residual second moments show why a combination can help. On path radiance, the residuals of the isotropic kernel and of the five-network ensemble, each divided by the norm of the true spectrum, have correlation 0.27 ± 0.10, below their error ratio of about 0.34 at every split; for a quadratic loss this is the classical condition under which a convex combination improves on its better member [81, 12]. The stack minimizes a mean relative norm instead, and the reflectance is a nonlinear function of the components, so the condition is suggestive rather than decisive for those metrics. The two residual-corrected predictors are far more strongly correlated, about 0.88 on path radiance and 0.995 on albedo.

Spherical albedo. Albedo has the largest relative error for every family: its small values in the short-wave infrared make relative errors sensitive to small absolute ones. The width-512 stack reaches 1.94%. The learning curves of Section 7 show little improvement with sample size on this component; Section 6 tests whether the out-of-domain training targets contribute to that plateau.

5.1 Paired comparisons and the radiance–reflectance trade-off

Paired differences compare two families on the same split. Table 2 reports them for the main and wide pipelines, with positive gains favoring the candidate. The wide feature kernel improves on the input-scaled kernel at all ten splits, by 0.0165 ± 0.0066 radiance percentage points. The wide stack's improvement over its own feature kernel is 0.0006 ± 0.0026 points, with eight wins and two losses: small against the split-to-split variation of the difference itself.

Radiance and reflectance order the families differently. Table 3 reports the all-band reflectance tails beside the medians; each entry averages a within-split statistic, so the mean of ten 95th percentiles is not a pooled 95th percentile. For the width-512 pipeline the convex stack lowers the radiance error relative to the feature kernel but raises the mean 95th percentile of the reflectance error from 7.75 to 29.44 points, and the tail is worse at all ten splits, by factors between 2.63 and

candidate / reference	mean gain ± SD [pp]	gain range [pp]	wins/10
Input-scaled / isotropic KRR	0.1436±0.0122	[0.1196, 0.1585]	10
Residual correction / network	0.2464±0.0090	[0.2365, 0.2615]	10
Narrow stack / input-scaled KRR	0.0083±0.0066	[-0.0054, 0.0185]	9
Wide feature kernel / input-scaled KRR	0.0165±0.0066	[0.0054, 0.0295]	10
Wide / narrow feature kernel	0.0191±0.0079	[0.0096, 0.0352]	10
Wide stack / wide feature kernel	0.0006±0.0026	[-0.0053, 0.0037]	8

Table 2: Paired radiance contrasts on seeds 101–110. Gain is the reference error minus the candidate error, in percentage points (pp). SD is the sample standard deviation of the ten paired differences; the range and the win count describe the same ten differences.

model	radiance [%]	median [pp]	95th percentile [pp]	RMSE [pp]
Cubic ridge	0.760±0.011	0.471±0.010	70.68±0.02	4.18e+03
Isotropic KRR	0.239±0.016	0.141±0.018	69.53±0.42	4.18e+03
Input-scaled KRR	0.095±0.009	0.036±0.004	31.16±6.21	1.34e+04
Narrow feature kernel	0.098±0.007	0.054±0.004	7.75±1.13	1.78e+03
Narrow convex stack	0.087±0.006	0.048±0.003	29.44±5.89	1.25e+03
Concatenated features, 5×512	0.076±0.007	0.039±0.004	5.93±0.47	1.06e+04
Wide feature kernel	0.079±0.009	0.045±0.006	7.57±0.95	1.40e+03
Wide convex stack	0.078±0.011	0.043±0.005	22.71±4.42	5.81e+03

Table 3: All-band retrieval metrics over the ten splits of the main, concatenated-feature and wide pipelines. Radiance is the mean relative L^2 error; the reflectance median, 95th percentile and RMSE are in percentage points at $\rho = 0.7$. The first three columns show mean ± sample standard deviation across splits and RMSE its mean. The inverse is neither clipped nor restricted to the physical domain; every inverse in these runs is finite, and the RMSE is dominated by large finite values near zeros of the denominator. Section 6 gives the same comparison on the physical domain.

4.94. The width-2000 pipeline repeats the reversal: 7.57 points for the feature kernel and 22.71 for the stack, worse at all ten splits by factors between 2.36 and 3.79. The near-tie at 0.079% against 0.078% radiance error hides a threefold difference in the inverse.

The input-scaled kernel shows the same split between center and tail: its median error is only 0.0365 points but its mean 95th percentile is 31.16 points, while the feature kernels have slightly larger medians and much smaller tails. The kernel on concatenated features has the smallest tail of all, 5.93 points. The identities of Section 3.1 tie the inverse error to the predicted denominator as well as to the forward error, which is why optimizing a component or radiance error alone need not control the reflectance tail.

6 The physical domain, the training targets and two tests fixed in advance

The all-band scores of Table 3 mix three things: the error of the emulator, entries where the retrieval is not defined, and entries where it is defined but badly conditioned. This section separates them. It describes the physical domain of the retrieval and the entries of the table

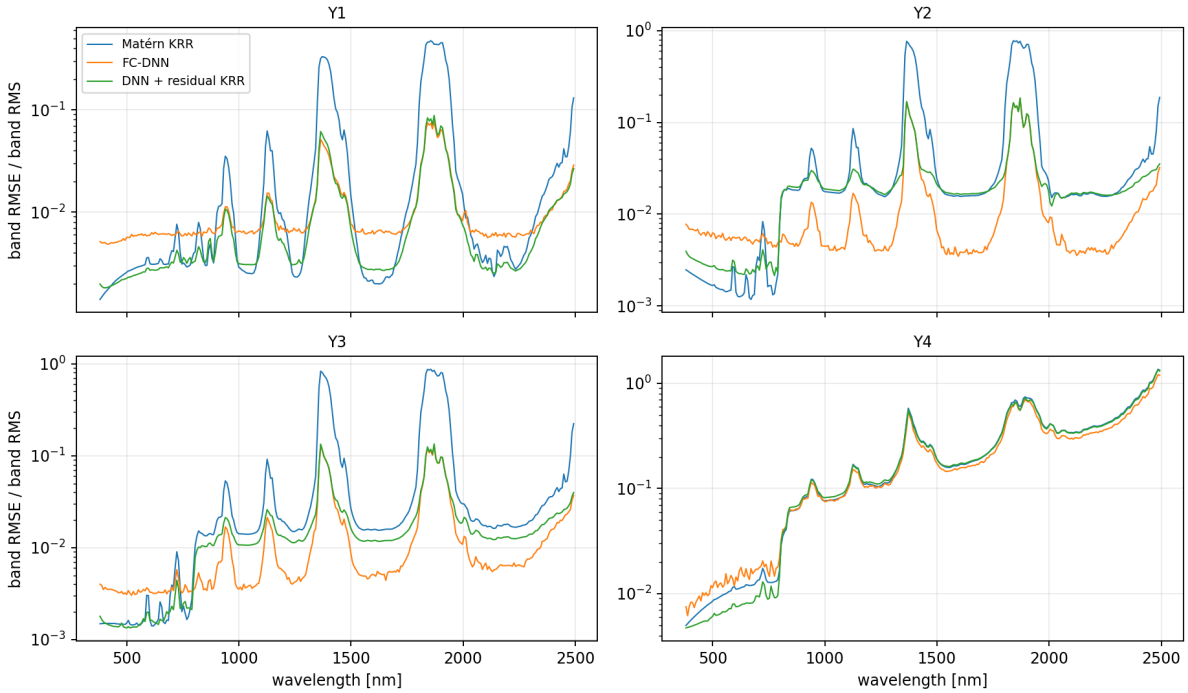


Figure 3: Where along the spectrum each family makes its error: band RMSE divided by the band's root-mean-square value over the test rows, against wavelength, for the isotropic Matérn kernel, the 3×512 network and the network with its residual kernel. The kernel's error follows the band structure, rising by two orders of magnitude in the water-vapour bands near 940, 1140, 1380 and 1870 nm; the network's is nearly flat across the spectrum and higher in the windows; the residual correction keeps the network's floor in the windows and most of the kernel's advantage outside them. On the spherical albedo Y_4 the three families agree to within a factor of two at every wavelength, and all of them degrade towards 2500 nm. The bandwise relative RMSE is a different aggregate from the per-sample relative norm of Table 1.

that fall outside it, scores the main pipeline on that domain with its coverage and failures, asks whether the out-of-domain entries matter when they are used as training targets, and repeats the forward–inverse comparison on a fresh partition and on a second radiative-transfer code, both fixed in advance.

6.1 The physical domain

Write $t = Y_2 + Y_3$, $s = Y_4$ and $q = 1 - \rho s$, with $\rho = 0.7$. The retrieval analysis of Section 9 holds on

$$\mathcal{A} = \{t > 0,\ 0 \le s < 1\},$$

which is a sufficient domain for the physical retrieval rather than the largest set on which the rational formulas are defined: a negative albedo with $t > 0$ gives $q > 0$ and a well-defined inverse, and $q < 0$ violates the physical hypothesis without being a zero of a denominator. At $q = 0$ the forward formula is undefined, and at $t = 0$, wherever it is defined, $L = Y_1$ does not depend on the reflectance, so the band carries no information about it. A small positive transmission makes

condition on the tabulated truth	entries	share
transmitted flux $t = Y_2 + Y_3 \le 0$	438	0.0066%
$t < 10^{-12}$	32,130	0.4836%
albedo $Y_4 < 0$	9,858	0.1484%
albedo $Y_4 \ge 1$	454	0.0068%
albedo exactly 2 (reported fill marker)	214	0.0032%
admissible $\mathcal{A} = \{t > 0,\ 0 \le Y_4 < 1\}$	6,633,849	99.8441%

Table 4: The table against the physical domain, over all $23{,}313 \times 285 = 6{,}644{,}205$ state–band entries. The categories overlap. Small positive transmission is a conditioning property, not a violation of $t > 0$; an albedo outside $[0, 1)$ violates the physical domain without necessarily leaving the algebraic domain of the inverse.

the inverse ill-conditioned but not undefined.

Table 4 counts the entries of the table outside $\mathcal{A}$. There are 438 nonpositive transmissions, the smallest -4.5×10^{-20}, and 32,130 transmissions below 10^{-12}; 9,858 negative albedos, the smallest -0.0724; and 454 albedos at or above one, 214 of which equal 2 exactly, which can only be a fill value. At 448 entries $q \le 0$, the smallest -0.4. In all, 10,356 entries (0.156%) lie outside $\mathcal{A}$. They occupy 227 bands but are concentrated near 1841–1908 and 2485–2493 nm, and 18,859 of the 23,313 states are admissible at every band.

A stricter domain adds a conditioning floor,

$$\mathcal{C} = \mathcal{A} \cap \{t \ge \tau,\ q \ge \kappa\}, \qquad \tau = 10^{-12}, \quad \kappa = 0.3,$$

with τ in the flux units of the table; at $\rho = 0.7$ the albedo condition already gives $q > 0.3$. Inverting the exact tabulated components returns the reflectance to within floating-point error, and that error is informative:

domain	round-trip RMSE (reflectance)	coverage
all entries	7.38×10^{-3}	100%
$\mathcal{A}$	4.27×10^{-3}	99.844%
$\mathcal{C}$	1.08×10^{-9}	99.516%

In exact arithmetic the round trip is exact wherever it is defined. In floating point, adding $\rho t/q$ to Y_1 and subtracting Y_1 again loses the added term when it is small relative to the spacing of numbers near Y_1, which is what happens at the smallest transmissions. Removing 0.33% of the entries takes the round-trip error from 4×10^{-3} to 10^{-9}; what remains of the all-band reflectance tail on $\mathcal{C}$ belongs to the emulators.

6.2 The main pipeline on the physical domain

The families of the main comparison were refitted on the ten splits at both network widths with their float64 test predictions kept; these are the raw arm of the experiment in Section 6.3. Each family is scored on the test entries that satisfy

$$t \ge \tau_0 T_{\text{train}}, \qquad 0 \le s < 1, \qquad q \ge 0.3,$$

family	radiance [%]	all bands p_{95} [pp]	$\tau_0 = 10^{-12}$ failed [%]	p_{95} [pp]	$\tau_0 = 10^{-3}$ failed [%]	p_{95} [pp]	$\tau_0 = 10^{-2}$ failed [%]	p_{95} [pp]
cubic ridge (3×512)	0.760±0.011	70.68±0.02	4.36±0.11	44.91±1.08	1.44±0.04	22.18±0.30	0.37±0.01	14.70±0.18
isotropic kernel (3×512)	0.239±0.016	69.53±0.42	3.13±0.17	21.30±2.18	0.57±0.07	7.01±0.86	0.08±0.01	4.09±0.43
input-scaled kernel (3×512)	0.095±0.009	31.16±6.21	1.95±0.13	5.50±0.67	0.20±0.07	1.55±0.16	0.02±0.01	1.06±0.10
network (3×512)	0.387±0.012	64.00±1.70	2.49±0.21	9.94±1.21	0.10±0.01	3.49±0.15	0.00±0.00	2.05±0.09
network + residual kernel (3×512)	0.141±0.006	47.29±4.40	2.09±0.08	6.90±0.33	0.05±0.00	2.33±0.13	0.00±0.00	1.44±0.07
kernel on features (3×512)	0.098±0.007	7.75±1.13	1.36±0.07	2.66±0.18	0.00±0.00	0.96±0.06	0.00±0.00	0.69±0.04
convex stack (3×512)	0.087±0.006	29.44±5.89	1.89±0.14	4.55±0.35	0.01±0.01	1.35±0.11	0.00±0.00	0.82±0.05
cubic ridge (3×2000)	0.760±0.011	70.68±0.02	4.36±0.11	44.91±1.08	1.44±0.04	22.18±0.30	0.37±0.01	14.70±0.18
isotropic kernel (3×2000)	0.239±0.016	69.53±0.42	3.13±0.17	21.30±2.18	0.57±0.07	7.01±0.86	0.08±0.01	4.09±0.43
input-scaled kernel (3×2000)	0.095±0.009	31.16±6.21	1.95±0.13	5.50±0.67	0.20±0.07	1.55±0.16	0.02±0.01	1.06±0.10
network (3×2000)	0.749±0.119	66.32±3.56	2.45±0.89	20.28±13.29	0.27±0.18	5.63±1.00	0.01±0.01	3.43±0.71
network + residual kernel (3×2000)	0.160±0.016	48.57±4.32	2.15±0.07	7.57±0.85	0.07±0.03	2.51±0.35	0.00±0.00	1.52±0.22
kernel on features (3×2000)	0.079±0.009	7.56±0.95	1.38±0.06	2.48±0.18	0.00±0.00	0.87±0.08	0.00±0.00	0.62±0.06
convex stack (3×2000)	0.078±0.011	22.71±4.43	1.80±0.10	3.77±0.34	0.00±0.00	1.12±0.11	0.00±0.00	0.70±0.06

Table 5: The main pipeline on the physical domain: the raw arm, mean ± sample standard deviation over ten splits. Radiance is the mean relative error in percent, and the all-band column is the unrestricted inverse of Table 3. At each floor τ_0, the failed column is the percentage of covered entries at which the inversion fails, and p_{95} is the 95th percentile of the absolute reflectance error over the successful inversions, in percentage points.

where T_{train} is the median positive transmission of the split's training block (4.67 at split 101) and $\tau_0 \in \{10^{-12}, 10^{-3}, 10^{-2}\}$; the three floors were fixed before the first fit, and at $\rho = 0.7$ the albedo condition already gives $q > 0.3$. An inversion fails when its predicted denominator is not positive or its value is not finite. Failures are counted, and the percentiles are taken over the successful inversions. Coverage depends on the truth alone and is the same for every family. The mask describes where the retrieval is well posed; it is an evaluation domain, not a filter that a retrieval could apply without knowing the true components.

On the physical domain every tail shrinks, and the order of the families does not change (Table 5). At the floor 10^{-12}, which keeps 99.3% of the test entries, the 95th percentile at width 512 runs from 2.66 points for the kernel on features to 44.9 for the cubic ridge, against 7.75 and 70.7 over all bands, and the stack's falls from 29.4 to 4.55. At 10^{-3} and 10^{-2}, with coverage 93.7–94.0% and 90.8–91.1%, the kernel on features and the stack stay below 1.4 points at both widths. The kernel on features has the smallest 95th percentile at every floor, on every split and at both widths. Failed inversions are few and sit at the smallest transmissions: at 10^{-12} they are 1.4 to 4.4% of the covered entries, and of those, between 69% for the cubic ridge and nearly all for the kernel on features and the stack lie below $10^{-3}T_{\text{train}}$; at 10^{-3} they are at most 1.4%, and at 10^{-2} at most 0.4%, almost all of them the cubic ridge's and the isotropic kernel's.

6.3 Training on physically admissible targets

The training pipeline standardizes, compresses and fits the targets as they come, including the 0.156% of entries outside $\mathcal{A}$ and the fill values among them. Scoring on $\mathcal{A}$ cannot tell whether those entries shaped the fitted representations, the principal components or the albedo plateau of Section 7; only refitting can. We therefore ran the pipeline under three training policies: the original one (*raw*); removal of every training state with an entry outside $\mathcal{A}$ at any band (*admissible*), with no clipping or imputation; and an unfiltered random subset of the raw training rows of the same size as the admissible set (*matched*), which separates the effect of the filter from that of losing rows. Whole states are removed so that every remaining spectrum is complete for the principal-component step; this changes the training distribution, and the matched arm

family	arm	Δ radiance [pp]	lower	Δ albedo [%]	Δ all-band p_{95} [pp]	Δ physical p_{95} [pp]	lower
cubic ridge (3×512)	admissible	0.0279±0.0022	0/10	-0.848±0.071	0.01±0.01	3.41±0.42	0/10
	matched	-0.0000±0.0015	5/10	0.134±0.271	0.00±0.01	-0.02±0.20	6/10
isotropic kernel (3×512)	admissible	0.0023±0.0149	7/10	-0.858±0.199	0.44±0.51	2.22±2.02	0/10
	matched	0.0112±0.0133	1/10	-0.033±0.316	0.02±0.27	0.58±1.53	4/10
input-scaled kernel (3×512)	admissible	-0.0068±0.0079	8/10	-0.668±0.178	7.26±7.24	1.29±0.76	0/10
	matched	0.0092±0.0118	2/10	0.062±0.258	5.73±9.01	0.75±1.40	3/10
network (3×512)	admissible	0.0258±0.0066	0/10	-1.219±0.203	3.47±1.25	1.84±1.60	2/10
	matched	0.0446±0.0115	0/10	-0.107±0.520	2.16±1.58	0.53±1.12	2/10
network + residual kernel (3×512)	admissible	0.0011±0.0056	4/10	-0.842±0.131	8.24±3.83	1.48±0.32	0/10
	matched	0.0126±0.0144	3/10	-0.200±0.312	4.31±3.27	0.74±0.35	0/10
kernel on features (3×512)	admissible	-0.0126±0.0069	10/10	-0.499±0.089	1.36±0.48	0.28±0.12	0/10
	matched	0.0042±0.0087	3/10	-0.105±0.211	1.01±1.60	0.26±0.31	1/10
convex stack (3×512)	admissible	-0.0085±0.0058	10/10	-0.451±0.070	3.83±5.39	0.48±0.33	1/10
	matched	0.0054±0.0089	3/10	-0.095±0.197	3.27±5.64	0.42±0.58	1/10
cubic ridge (3×2000)	admissible	0.0279±0.0022	0/10	-0.848±0.071	0.01±0.01	3.41±0.42	0/10
	matched	-0.0000±0.0015	5/10	0.134±0.271	0.00±0.01	-0.02±0.20	6/10
isotropic kernel (3×2000)	admissible	0.0023±0.0149	7/10	-0.858±0.199	0.44±0.51	2.22±2.02	0/10
	matched	0.0112±0.0133	1/10	-0.033±0.316	0.02±0.27	0.58±1.53	4/10
input-scaled kernel (3×2000)	admissible	-0.0068±0.0079	8/10	-0.668±0.178	7.26±7.24	1.29±0.76	0/10
	matched	0.0092±0.0118	2/10	0.062±0.258	5.73±9.01	0.75±1.40	3/10
network (3×2000)	admissible	0.0107±0.2159	5/10	-1.239±0.442	2.75±3.52	-5.11±9.97	8/10
	matched	0.0505±0.1926	3/10	0.016±0.407	1.64±4.91	-5.89±14.21	7/10
network + residual kernel (3×2000)	admissible	-0.0091±0.0216	6/10	-0.784±0.179	5.85±7.87	0.92±1.69	4/10
	matched	0.0146±0.0203	3/10	-0.035±0.242	3.90±7.14	0.60±1.28	4/10
kernel on features (3×2000)	admissible	-0.0163±0.0090	10/10	-0.609±0.342	0.65±1.74	0.11±0.47	5/10
	matched	0.0015±0.0156	5/10	-0.208±0.286	1.39±3.97	0.27±0.70	4/10
convex stack (3×2000)	admissible	-0.0171±0.0105	10/10	-0.509±0.239	-1.97±5.61	-0.13±0.63	6/10
	matched	0.0002±0.0149	6/10	-0.132±0.250	2.14±8.91	0.29±1.03	4/10

Table 6: Training on admissible targets: paired differences from the raw arm, arm minus raw, mean ± sample standard deviation over ten splits. Radiance differences are in percentage points of relative error, albedo differences in percentage points of the relative albedo error, and the reflectance percentiles in percentage points; the physical p_{95} uses the floor $\tau_0 = 10^{-12}$. The columns headed lower count the splits on which the arm is below the raw arm.

measures how much of any difference is only the loss of 19% of the rows. The validation rows, the validation loss and the test blocks are those of the raw pipeline in all three arms, and every arm is scored on the same test entries with a flux scale taken from the original training block.

The arms, the two network configurations of Table 1 and Table 13, the seeds 101–110, the scoring and the summaries reported below were fixed in a written protocol before the first run, and every run was completed and is reported. Each run records the digests of its data and of its split indices, and saves the float64 predictions of every family. Splits 101–105 ran on one machine and splits 106–110 on another, and the three arms of a split always ran on the same machine, so every paired difference below is taken within one computing environment (Table S6).

The filter is not neutral. The states it removes, 18.8 to 19.3% of each training block, are wet, low and oblique: at split 101 their median water-vapour column is 5.1 cm against 3.0 cm for the states kept, their median elevation 1.1 km against 3.4 km and their median solar zenith angle 41.7° against 30.5°, and the smallest transmission of the median removed state is 2×10^{-15} against 1×10^{-5}. On every split they hold a third of the training entries with transmission below $10^{-3} T_{\text{train}}$ (32.7 to 33.4%). In the test blocks 18 to 20% of the states are of the kind the filter removes, and their entries inside the physical domain are scored like any others.

Training on admissible targets improves the components and not the retrieval (Table 6). The

width	arm	stack − features radiance [pp]	lower	tail ratio all bands	> 1	tail ratio $\tau_0 = 10^{-12}$	tail ratio $\tau_0 = 10^{-3}$	tail ratio $\tau_0 = 10^{-2}$	> 1
3×512	raw	-0.0109±0.0024	10/10	3.84±0.77	10/10	1.71±0.09	1.40±0.08	1.19±0.06	10/10
	admissible	-0.0068±0.0014	10/10	3.69±0.47	10/10	1.71±0.10	1.47±0.02	1.24±0.03	10/10
	matched	-0.0098±0.0023	10/10	3.79±0.89	10/10	1.71±0.17	1.44±0.10	1.23±0.05	10/10
3×2000	raw	-0.0006±0.0026	8/10	3.00±0.40	10/10	1.52±0.08	1.29±0.07	1.13±0.05	10/10
	admissible	-0.0014±0.0006	10/10	2.55±0.26	10/10	1.41±0.08	1.26±0.04	1.12±0.02	10/10
	matched	-0.0019±0.0017	9/10	2.82±0.51	10/10	1.47±0.08	1.29±0.06	1.13±0.04	10/10

Table 7: The convex stack against the kernel on features in each arm: the radiance difference, stack minus feature kernel, in percentage points, and the ratio of their 95th-percentile reflectance errors over all bands and on the physical domain at the three floors; mean ± sample standard deviation over ten splits. The column headed lower counts the splits on which the stack has the lower radiance error, and the columns headed > 1 the splits on which it has the larger tail, over all bands and at $\tau_0 = 10^{-12}$.

albedo error falls for every family, by 0.45 to 1.22 points at width 512 and 0.51 to 1.24 at width 2,000. On radiance the kernel on the network's features and the stack gain on every split, by 0.013 and 0.009 points at width 512 and 0.016 and 0.017 at width 2,000. The network itself loses 0.026 points on average at width 512, on every split, so there, as with the wider network of Section 7, the result is a worse emulator and a better feature map; at width 2,000 the network is worse on five of the ten splits and 0.011 points worse on average. The gain comes from the filter and not from the smaller training set: the matched arm raises the feature kernel's radiance error at both widths. At width 2,000 the stack reaches 0.061%, against 0.078% when trained on every row. The physical tail moves the other way. At width 512 the 95th percentile at the floor 10^{-12} rises for every family on average, by 0.28 to 3.41 points, and for each family on at least eight of the ten splits; at width 2,000 it rises on every split for the three families without a network and is mixed for the four built on one. The matched arm separates the causes. For the cubic ridge, the isotropic kernel and the network, the random subset of the same size raises the tail by at most 29% of the filter's rise, so the filter is what raises it; for the kernel on features it raises it by 92% of the filter's rise, so there the cause is the loss of rows; the input-scaled kernel, the corrected network and the stack lie between, at 50 to 87%.

Figure 4 shows where the filter moves the error at width 512. Relative to the raw arm, the admissible arm's conditional mean square $\mathbb{E}[e_R^2 \mid t]$ is larger in each of the three bins of the lowest tenth of the transmissions for every family, by factors up to 2.8 and, for the input-scaled kernel, 4.7 in the lowest percent, and smaller in the top half for every family, by factors of 0.17 to 0.79; in between, the ratio crosses one. The matched arm has no such tilt: its ratios lie between 0.70 and 1.38 in every bin for every family but the input-scaled kernel, which is worse by 1.2 to 2.6 throughout. Removing the wet, low and oblique states moves error from the transmissions that dominate the forward metric to those that dominate the retrieval, which is the trade that Theorem 6 prices: the profile of the error in t, not its mean, sets the tail. At width 2,000 the network does not show the tilt in the lowest bins, in line with the mixed tails there.

The reversal survives all three policies (Table 7). In every arm the stack has the larger 95th percentile on every split, over all bands and at every floor: at width 512 in the raw arm by factors of 2.63 to 4.94 over all bands, 1.56 to 1.87 at 10^{-12} and 1.10 to 1.30 at 10^{-2}, while its radiance error is lower on every split. At width 2,000 the two are level on radiance in the raw arm, the stack lower on eight of ten splits by 0.0006 points on average, and the tail ratio stays

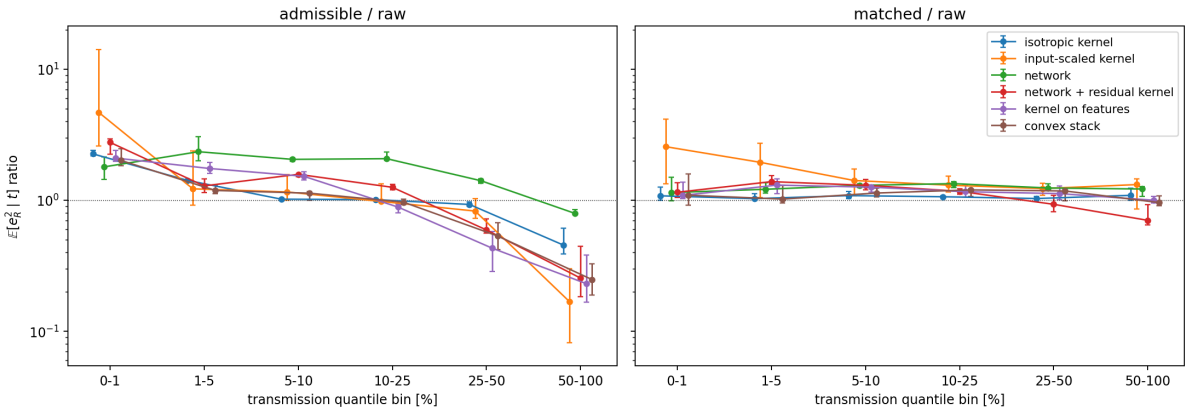


Figure 4: Where the admissible targets move the error at width 512: the ratio of the conditional mean square $\mathbb{E}[e_R^2 \mid t]$ in each arm to that of the raw arm, in six quantile bins of the true transmission on the physical domain (0–1, 1–5, 5–10, 10–25, 25–50 and 50–100 percent), one line per family; points are medians over the ten splits and bars the interquartile range. The admissible arm raises the error at small transmissions and lowers it in the top half; the matched arm, which loses the same number of rows at random, has no such tilt. This analysis is exploratory and was not in the written protocol.

above one everywhere. The ratio shrinks as the floor rises, which is what Theorem 4 leads one to expect: the stack's excess error sits at the smallest transmissions.

6.4 Where the reflectance error sits

Figure 5 shows where along the spectrum the errors of one split sit. The component and radiance residuals are smooth in wavelength and widen where the transmitted flux is small, so the families differ mainly in the amplitude of a shared shape. The reflectance residual does not inherit that shape. Over all bands its outer envelopes open into vertical walls, bands where the 95th percentile of the absolute error exceeds two points, and the walls have two parts. A common core of 54 bands, at 1126–1134, 1342–1469, 1796–1952 and 2412–2493 nm, is a wall for every family, the most accurate included: there the denominator, not the emulator, sets the width. Outside the core each family adds bands in proportion to its inaccuracy: none for the kernel on features, 3 for the stack, 12 for the input-scaled kernel, 22 for the corrected network, 29 for the network and 79 for the cubic ridge, of a 133-band union. Restricting to the physical domain changes none of these counts, and neither does the stricter floor $\tau_0 = 10^{-3}$, which removes 6.3% of the entries: the walls come from transmissions that are small but far above either floor, where the retrieval divides a small component error by a small denominator. The feature kernel's advantage in the tail is that it adds no wall of its own.

Theorem 4 locates the walls in advance. Its item (i) holds at every entry, so at each band the 95th percentile of the constrained retrieval error is at most the 95th percentile of $\min\{R, e_R/t\}$, which uses only the component errors and the true transmission. On split 101, with $R = 0.9$ and S the largest training albedo not above one, the constrained retrieval on the physical domain has exactly the walls of Figure 5, family by family, and no entry violates the pointwise inequality. For every family each wall lies in the set of bands where the bound's 95th percentile exceeds two points; that set is larger by 8 to 21 bands, and on the walls the bound exceeds the realized 95th

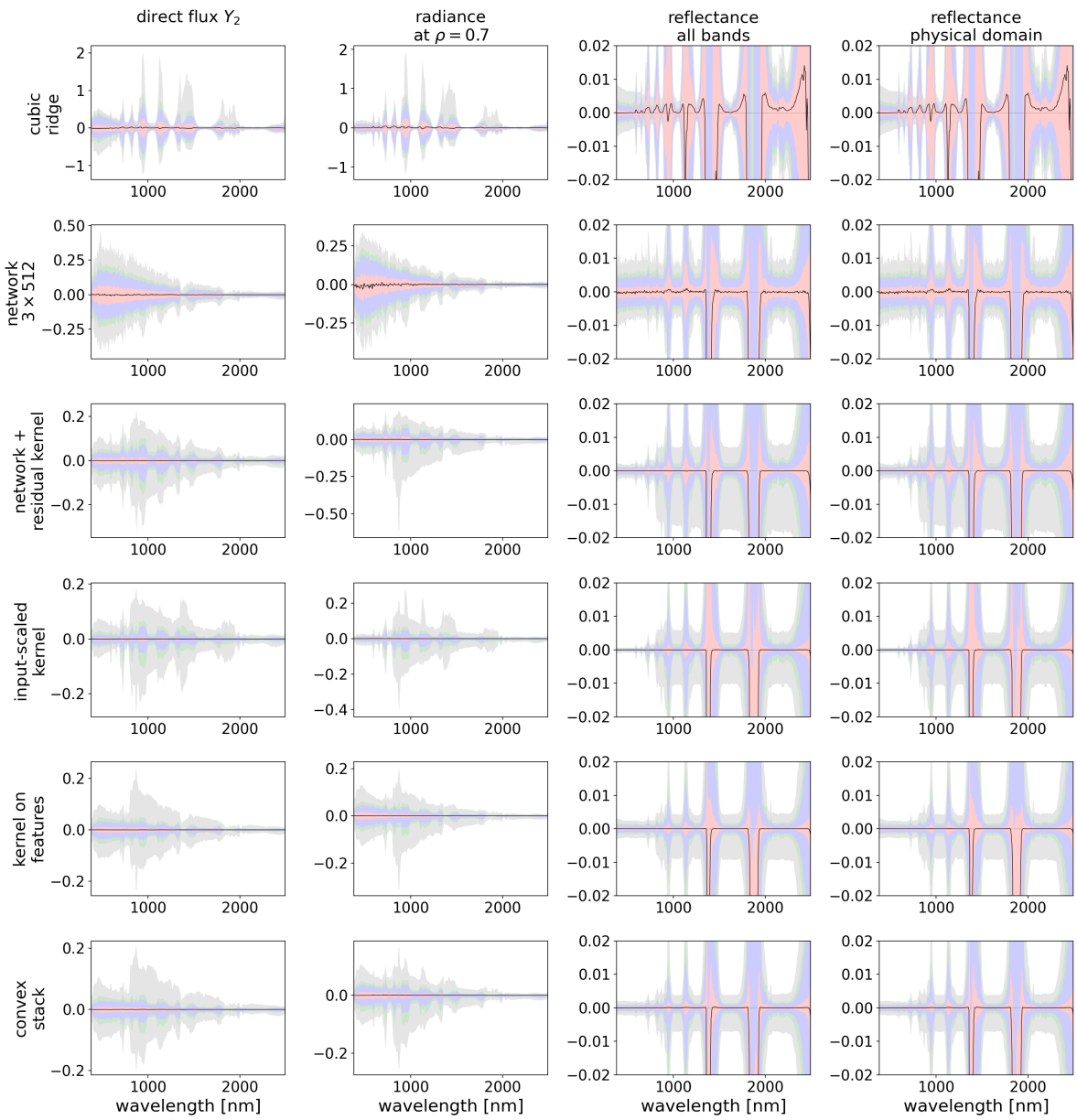


Figure 5: Residual quantile bands across the test set of split 101, band by band, for six families of the main pipeline. The envelopes are nested percentiles of the residual at each wavelength: 25–75 in red, 5–25 and 75–95 in blue, 2.5–5 and 95–97.5 in green, 0.5–2.5 and 97.5–99.5 in grey, with the median as a thin black line. From the left: the direct-flux residual $Y_2 - \widehat{Y_2}$ in physical units; the radiance residual at $\rho = 0.7$; the reflectance residual over all bands; and the reflectance residual on the physical domain with the floor $\tau_0 = 10^{-12}$. The last two columns share a vertical scale of ± 2 points.

family	radiance rel. L_2 [%]	all bands [pp]			screened [pp]	
		median	p_{95}	p_{99}	median	p_{95}
cubic ridge	0.7544	0.475	70.72	129.1	0.468	70.67
network 3×512	0.1719	0.106	41.83	84.1	0.105	28.07
network + residual kernel	0.0978	0.061	26.07	80.5	0.061	16.87
input-scaled Matérn kernel	0.0900	0.028	26.89	189.4	0.027	19.38
kernel on learned features	0.0828	0.034	3.91	129.2	0.034	2.94
convex stack	0.0687	0.044	25.82	89.6	0.044	16.30

Table 8: The fresh partition: 3,497 held-out states, evaluated once. The screened columns use $t \geq 10^{-12}$ and $q \geq 0.3$, which covers 99.5327% of the 996,645 entries; no inversion is non-finite in any family. The network here is the 3×512 GELU network of the frozen pipeline, trained with Adam, not the network of Table 1.

percentile by a median factor of 1.30 to 1.38, depending on the family, and never by more than 4.5. The bands marked by the bound for all six families number 64 and include the whole common core of 54. The same holds on every split at both widths: no entry violates the inequality, every wall of every family lies in the marked set, which is larger by 4 to 85 bands, and on the walls the bound exceeds the realized 95th percentile by median factors of 1.29 to 1.43.

6.5 A fresh partition, fixed in advance

The comparisons above use partitions that were in view while the families were developed. To test the reversal against that selection, a simplified pipeline was frozen together with one hypothesis in a written protocol, the SHA-256 of its hyperparameters was recorded, and the pipeline was then evaluated once on a new partition. Its six families are the cubic ridge, a 3×512 network (GELU activations, Adam) trained on the 285 standardized bands, the residual kernel of that network, the input-scaled Matérn kernel, the Matérn kernel on the network's last hidden layer, and a convex stack of the cubic, the corrected network, the input-scaled kernel and the feature kernel. Every hyperparameter was copied from development on seed 101 and none was re-tuned. The partition holds out 3,497 states, 15% of the table, drawn with a seed that defines no other split; the remaining 19,816 states train the families, 1,982 of them serving only as the network's early-stopping set. The held-out states belong to the same table as every other split, so this evaluation guards against the choice of split and of hypothesis rather than testing generalization to new states.

The hypothesis was that the family with the lowest radiance error does not have the lowest all-band 95th-percentile reflectance error, and that its all-band 95th percentile is strictly larger than that of the residual-corrected network.

The reversal reproduces (Table 8). The convex stack has the lowest radiance error of the six, 0.0687%, and the second-lowest reflectance tail; the kernel on learned features has a larger radiance error, 0.0828%, and an all-band 95th percentile smaller by a factor of 6.60, and by 5.54 on the screened entries. Both factors are close to the 6.13 and 4.99 of the same pipeline on its development split, seed 101. The second clause of the hypothesis fails: the stack's all-band 95th percentile is 25.82 points and the corrected network's 26.07, smaller by 0.25 points where the hypothesis required it to be larger. The qualitative statement survives a single read of a fresh partition, while the stack, the corrected network and the input-scaled kernel, whose tails lie

family	radiance [%]	all bands p_{95} [pp]	$\tau_0 = 10^{-12}$ failed [%]	p_{95} [pp]	$\tau_0 = 10^{-3}$ failed [%]	p_{95} [pp]	$\tau_0 = 10^{-2}$ failed [%]	p_{95} [pp]
cubic ridge	0.986±0.037	54.17±1.23	3.28±0.13	4.47±0.38	0.00±0.00	2.71±0.11	0.00±0.00	2.44±0.08
isotropic kernel	0.639±0.026	59.90±1.88	3.16±0.12	3.91±0.37	0.00±0.00	1.93±0.12	0.00±0.00	1.69±0.11
input-scaled kernel	0.172±0.010	11.42±2.08	1.13±0.15	3.42±0.38	0.00±0.00	0.46±0.05	0.00±0.00	0.41±0.04
network	0.359±0.015	19.62±2.27	2.22±0.12	2.31±0.23	0.00±0.00	0.85±0.06	0.00±0.00	0.77±0.05
network + residual kernel	0.318±0.015	19.68±2.69	2.20±0.08	2.27±0.27	0.00±0.00	0.81±0.07	0.00±0.00	0.72±0.06
kernel on features	0.188±0.007	10.31±1.13	1.57±0.08	2.30±0.18	0.00±0.00	0.49±0.03	0.00±0.00	0.44±0.02
convex stack	0.182±0.008	10.42±1.27	1.54±0.11	2.34±0.26	0.00±0.00	0.48±0.03	0.00±0.00	0.43±0.03

Table 9: The pipeline on libRadtran, mean ± sample standard deviation over ten splits, with the columns of Table 5.

within about a point of one another, do not separate at this sample size.

6.6 A second radiative-transfer code

Every comparison so far concerns one simulator and one table. The corpus of [4] pairs 6S [14] with libRadtran [15, 16] over the thirteen Sentinel-2 bands and gives, for each atmospheric state and band, the path reflectance ρ_{path}, the total transmittance T and the spherical albedo S computed by libRadtran. The top-of-atmosphere reflectance is $\rho_{\text{path}} + \rho T/(1 - \rho S)$, which is (1) with $Y_1 = \rho_{\text{path}}$, $t = T$ and $Y_4 = S$, so the retrieval (2), the physical domain and Theorem 4 apply without change. We use the 9,722 of its 50,000 states that carry all thirteen bands; the others lack band B10 at 1372 nm, which lies in a strong water-vapour absorption band and is nearly opaque. Its median transmittance is 9×10^{-6} and a third of its entries lie below 10^{-6}, while the first percentile of every other band is above 0.2. The inputs are the seven continuous state variables (solar and view zenith, relative azimuth, aerosol optical depth, water vapour, ozone and elevation) and the aerosol model and atmosphere profile of the state, one-hot encoded with four and five levels; the transmission enters as $Y_2 = T$ with a negligible $Y_3 = 10^{-6}T$, so that the pipeline runs unchanged. The two categorical inputs carry much of the variation: fitted within each of the twenty aerosol-profile classes, a cubic regression of the log transmittance on the continuous inputs predicts the transmittance at bands B2, B4, B8A and B11 with a root-mean-square relative error of 0.7–3.0%, against 2.1–8.7% for one fit across the classes. The same within-class fit reproduces the 6S transmittance of the same states to 0.1–0.2%, so part of libRadtran's variation is not a smooth function of these inputs.

The pipeline of Section 4 was run on these arrays with no change to any family or hyperparameter grid, at output rank 13 (all bands), on ten splits of the same form (7,875 training, 875 validation and 972 test states), and scored as in Section 6.2. None of these states played a part in developing the families. Three hypotheses and their decision rule were fixed in a written protocol, amended once, to add the categorical inputs, before any run reported here was fitted: (H1) the family with the lowest radiance error does not have the lowest all-band 95th-percentile reflectance error; (H2) the convex stack is no worse than the kernel on features in radiance and has the larger all-band tail; (H3) restricting the evaluation to transmissions above $10^{-3}T_{\text{train}}$ brings the families' reflectance tails into closer rank agreement with their radiance errors, measured by Kendall's τ. Each was to be accepted if it held on at least nine of the ten splits.

Table 9 gives the result. H1 holds on six of the ten splits and is not confirmed: the lowest radiance error belongs to the input-scaled kernel on eight splits and to the convex stack on two splits, and the lowest all-band tail to the convex stack on four splits, to the input-scaled kernel on three splits and to the kernel on features on three splits. H2 holds on five of the ten splits and



Figure 6: The 95th percentile of the constrained retrieval error on libRadtran at $\rho = 0.7$, band by band and family by family, in percentage points, over the test states of all ten splits; the colour scale is logarithmic. At B10 the retrieval of every family returns zero on at least five percent of the states.

is not confirmed: the stack is no worse than the kernel on features on radiance on every split and has the larger all-band tail on five of the ten splits, the ratio of the two 95th percentiles ranging from 0.92 to 1.13. H3 holds on nine of the ten splits and is confirmed: Kendall's τ between the radiance and tail rankings, 0.52 to 0.90 over all bands, rises to 0.81 to 1.00 above $10^{-3}T_{\text{train}}$.

One band sets the all-band tail (Figure 6). At B10 every family loses the retrieval: its 95th percentile is 70 points, the error of a constrained retrieval that returns zero at $\rho = 0.7$. On the other twelve bands the largest per-band 95th percentile is 0.56 to 3.44 points, and band by band the stack and the kernel on features differ by at most 0.03 points. Away from B10 every transmission exceeds 0.14, so the factor $1/t$ in Theorem 4 stays below 8, whereas on the EMIT table 5.5% of the test entries have transmissions between 10^{-12} and 10^{-3} times T_{train}. Without such transmissions Theorem 4 bounds the retrieval error at every entry by about seven times the retrieval-weighted component error. The stack's all-band tail is 0.92 to 1.13 times that of the kernel on features, against 2.63 to 4.94 on the EMIT table at width 512, and since the two differ by at most 0.03 points on the other bands, the difference comes from B10.

7 Learning curves, output rank and network width

The experiments below vary training size, output rank and network width under the same validation-based selection. Learning curves include cubic ridge, the isotropic and input-scaled kernels, the network, its residual correction and the feature kernel.

Learning curves. Each rung uses the first n rows of the split's training block, $n = 500$, 1,000, 2,000, 4,000, 8,000 and 18,884, with the validation and test blocks unchanged, so the rungs of one split are nested (Table 10, Figure 7). The cubic error changes little, from 0.85% to 0.76% over a factor of 38 in data, and most of that decrease is reached by a thousand rows. On the first three components every other family decays at close to $n^{-1/2}$: the least-squares slopes of log error against $\log n$ over the six rungs lie between -0.38 and -0.61 for the kernel, the network,

training rows	seeds	Cubic ridge	KRR, isotropic	KRR, per-input	FC-DNN	DNN + residual KRR	kernel on features
500	10	0.846±0.031	0.639±0.022	0.415±0.024	1.961±0.057	0.815±0.035	0.456±0.020
1,000	10	0.802±0.024	0.519±0.034	0.260±0.066	1.862±0.015	0.590±0.027	0.316±0.040
2,000	10	0.780±0.016	0.424±0.051	0.217±0.052	1.232±0.034	0.376±0.041	0.240±0.049
4,000	10	0.769±0.010	0.328±0.028	0.154±0.037	0.830±0.019	0.256±0.021	0.164±0.033
8,000	10	0.763±0.010	0.279±0.020	0.119±0.027	0.584±0.020	0.190±0.014	0.121±0.017
18,884	10	0.760±0.011	0.239±0.016	0.095±0.009	0.387±0.012	0.141±0.006	0.098±0.007

Table 10: Learning curves: radiance relative L^2 error [%] against the number of training rows, mean ± standard deviation over ten splits. Each rung uses the first n rows of the split's training block; the validation and test blocks are those of Table 1, so the top rung is that table. The per-input kernel repeats its coordinate search at every rung. Component errors are shown in Figure 7.

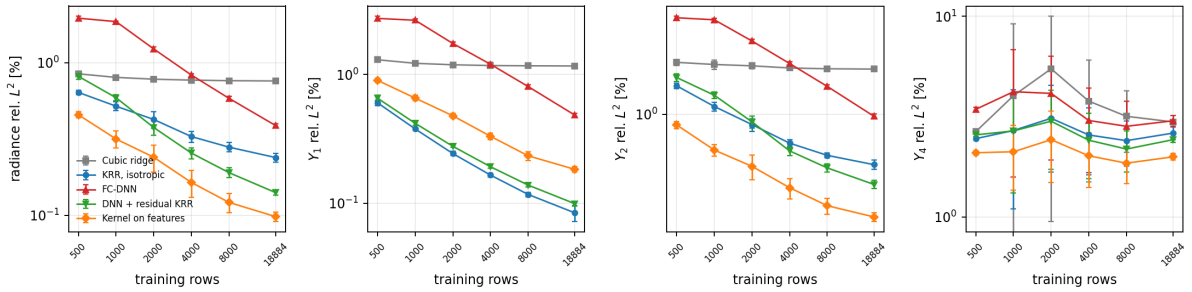


Figure 7: Learning curves on log–log axes for the radiance and for three of the four components, mean and standard deviation over ten splits. The cubic error changes little over the range. The kernel, network, residual-corrected network and feature kernel decrease approximately as $n^{-1/2}$ on the first three components, with fitted slopes in Table 11. The albedo error shows no sustained decrease; unusually large errors on individual splits widen its spread at 1,000 and 2,000 rows.

the corrected network and the feature kernel on Y_1, Y_2 and Y_3 (Table 11). On the albedo every slope lies between -0.09 and $+0.01$. These are slopes over a finite range of sample sizes; the rates of kernel regression theory require sampling, source and regularization conditions [47, 49] that this experiment does not check.

The isotropic kernel bends at the largest size. Its slope over the first five rungs is -0.59, -0.44 and -0.47 on the smooth components, and its last step is -0.38, -0.19 and -0.17, while the network's last step is as steep as its average (-0.59, -0.60 and -0.55). Part of the bend is the tuning protocol: the kernel's smoothness, scale and nugget are chosen on a 6,000-row subsample at every rung, so above 6,000 rows the hyperparameters belong to a smaller problem, and Section S5 shows that tuning with every row recovers much of the lost slope. The ratio of the network's radiance error to the kernel's falls from 3.1 at 500 rows to 1.6 at the full block, or to 1.8 with the kernel tuned on every row. The kernel's margin over the network is a margin at this training size.

The per-input metric changes the slope as well as the level. With its coordinate search repeated at every rung it has slope -0.399 on radiance against the isotropic kernel's -0.279, and the ratio of the two errors falls from 0.65 at 500 rows to 0.40 at the full block, with one step against that direction (0.50 at 1,000 rows, 0.51 at 2,000). The selected metric, smoothness and nugget all change with the sample size, so this is not a change of asymptotic dimension. Against the kernel on network features the margin closes as the data grow. The per-input metric has the

	slope over the six rungs					slope of the last step				
family	Y_1	Y_2	Y_3	Y_4	radiance	Y_1	Y_2	Y_3	Y_4	radiance
Cubic ridge	−0.03	−0.03	−0.03	−0.02	−0.03	−0.00	−0.01	−0.00	−0.08	−0.01
KRR, isotropic	−0.55	−0.38	−0.40	−0.01	−0.28	−0.38	−0.19	−0.17	0.10	−0.18
KRR, per-input	−0.45	−0.43	−0.43	0.01	−0.40	−0.32	−0.26	−0.29	0.12	−0.26
FC-DNN	−0.50	−0.49	−0.57	−0.09	−0.48	−0.59	−0.60	−0.55	0.07	−0.48
DNN + residual KRR	−0.52	−0.53	−0.61	−0.04	−0.50	−0.39	−0.33	−0.39	0.12	−0.35
Kernel on features	−0.45	−0.44	−0.46	−0.03	−0.43	−0.28	−0.22	−0.31	0.08	−0.25

Table 11: Learning-curve exponents: the least-squares slope of log error against $\log n$ over the six rungs of Table 10, and the slope of the last step alone, from 8,000 to 18,884 rows, per family and component.

PCA rank	variance outside the basis	KRR, isotropic	FC-DNN	DNN + residual KRR	kernel on features
16	7.5×10^{-4}	0.270±0.015	0.390±0.015	0.175±0.007	0.148±0.009
32	1.5×10^{-5}	0.243±0.018	0.390±0.015	0.139±0.006	0.102±0.009
64	8.2×10^{-8}	0.242±0.018	0.390±0.015	0.143±0.006	0.100±0.009
128	5.4×10^{-9}	0.242±0.018	0.390±0.015	0.150±0.005	0.100±0.009
285	0	0.242±0.018	0.390±0.015	0.172±0.004	0.100±0.009

Table 12: Output-rank ablation: radiance relative L^2 error [%] when the kernels and corrections regress the leading r principal coefficients, mean ± standard deviation over five splits. The second column is the fraction of the standardized variance the basis leaves out for the slowest-decaying component, the spherical albedo; at rank 285 it is zero. The network regresses the 285 bands directly and is the same predictor in every row.

lower mean at all six rungs, but it wins at 10, 9, 10, 7, 5 and 6 of ten splits from 500 rows to the full block, and the gap narrows from 0.415 against 0.456 at 500 rows to 0.095 against 0.098 at the top; the feature kernel's own slope, −0.435, is the steeper. On the albedo the per-input slope is +0.01.

The corrected network is below both of its parents on radiance at every split from 2,000 rows on, and at no split at 500 or 1,000 rows, where its error lies between the network's and the kernel's. On Y_2 and Y_3 it is below both at every split at the two largest sizes. On Y_1 the isotropic kernel has the lower mean at every size (0.084% against 0.099% at the full block), and the corrected network is below both parents at no split short of the full block and at two of ten there. Its exponent is close to the network's and its level close to the kernel's. Section 8 gives the exact residual decomposition behind this behavior.

Output rank. The kernels and corrections regress the leading r principal coefficients of the standardized component, while the network regresses the 285 bands and is the same predictor at every rank. Table 12 refits at $r = 16$, 32, 128 and 285 on five splits beside the rank-64 rows of those splits. The kernel changes little above rank 32: its radiance error reads 0.243, 0.242, 0.242 and 0.242 at 32, 64, 128 and 285, and 0.270 at 16, where the truncation leaves 7.5×10^{-4} of the albedo's variance outside the basis. The kernel on features behaves the same way. The residual correction does not: it is best at 32 (0.139), slightly worse at 64 (0.143), and worse again at 128 and 285 (0.150 and 0.172). The two regressions have different targets. The direct kernel fits the component, whereas the correction fits the projected network residual, and a higher rank exposes the correction to more directions of network error even where the target has little variance. The

network	head	component rel. L^2 [%]				radiance	reflectance
		Y_1	Y_2	Y_3	Y_4	rel. L^2 [%]	med. $\lvert\hat{\rho}-\rho\rvert$ [pp]
3×512	network	0.48±0.02	0.97±0.04	0.41±0.02	3.02±0.18	0.387±0.012	0.245±0.008
	network + residual KRR	0.10±0.00	0.30±0.02	0.16±0.01	2.43±0.08	0.141±0.006	0.089±0.008
	kernel on its features	0.18±0.01	0.17±0.01	0.10±0.01	1.99±0.08	0.098±0.007	0.054±0.004
	per-coordinate selection	0.19±0.01	0.44±0.10	0.13±0.02	2.25±0.18	0.193±0.022	0.123±0.023
	convex stack	0.07±0.00	0.16±0.01	0.09±0.00	1.94±0.06	0.087±0.006	0.048±0.003
3×2000	network	0.93±0.20	2.11±0.49	0.76±0.19	3.27±0.19	0.748±0.116	0.508±0.110
	network + residual KRR	0.12±0.01	0.35±0.03	0.19±0.03	2.35±0.13	0.160±0.016	0.106±0.010
	kernel on its features	0.13±0.02	0.13±0.02	0.08±0.01	2.10±0.34	0.079±0.009	0.045±0.006
	per-coordinate selection	0.14±0.02	0.30±0.26	0.09±0.01	2.55±0.26	0.187±0.057	0.099±0.033
	convex stack	0.08±0.01	0.13±0.02	0.08±0.00	2.00±0.24	0.078±0.011	0.043±0.005

Table 13: The pipeline of Section 4 with a 3×2000 network in place of the 3×512 one, on the same ten splits: the network (one member, up to 500 epochs under the same optimizer, learning rate and early stopping), its residual correction, the exact kernel on its last-layer features, and the per-coordinate selection and convex stack over the wide pipeline's four distinct predictors (the isotropic kernel, network, correction and feature kernel; with a single network the ensemble slots repeat the network), beside the corresponding rows of Table 1; mean ± standard deviation over the ten splits.

projection identity of Section 8 separates the two error terms.

Width. A network of three hidden layers of 2,000 units, trained for up to 500 epochs with the optimizer, learning rate and early stopping of the 512-unit network, was run on all ten splits with the rest of the pipeline built on it (Table 13). It is a worse emulator on every split, 0.75% against 0.39% on radiance, and early stopping fired between 42 and 210 of the 500 epochs allowed. Width and training schedule change together here, so this describes the trained networks rather than width as such. The exact kernel on its last-layer features reaches 0.079% on radiance against 0.098% on the 512-unit features; it is below the per-input kernel's 0.095% on all ten splits and below the stack of Table 1 on nine, and on the two flux components, at 0.13% and 0.08%, it is at or below every row of that table. The convex stack over the wide pipeline's heads reaches 0.078%, below the stack of Table 1 on nine of ten splits, with a median reflectance error of 0.043 points; the per-coordinate selection over the same heads again does not help. The residual correction of the wider network is slightly worse than that of the narrower one. A 3×1000 network on the same ten splits sits between the two: the network at 0.72%, the kernel on its features at 0.088%. From 512 to 2,000 units the network's own error rises with width while the error of a kernel on its features falls. At 4,000 units, on three splits, the network is worse again at 1.10% and the kernel on its features turns back up to 0.099%, so the gain in the representation stops by 2,000 units.

Concatenated features. If the hidden layer is a good kernel metric, several of them together may be better. On all ten splits we solved one exact kernel on the concatenated last-layer features of the five width-512 networks, 2,560 coordinates in all. It reaches 0.076 ± 0.007% on radiance against 0.098 ± 0.007% for the features of one network, lower on every split by 0.022 ± 0.003 points, and 0.039 against 0.054 points of median reflectance error. Its mean within-split 95th-percentile reflectance error is 5.93 points against 7.75. One kernel solve on concatenated features therefore beats the six-head stack of Table 1 on radiance, 0.076% against 0.087%, and has a far smaller reflectance tail (Table 3). On splits 104–107 the same construction with three width-2,000

networks, 6,000 coordinates, reaches 0.069% against 0.076% for a single member's features, lower on three of the four, and 0.033 against 0.040 reflectance points; there it is level with the six-head stack fitted on the same splits (0.069%), whose reflectance tail is more than twice as large (Table S1).

Concatenation averages metrics, where a stack or an ensemble averages predictions. After standardization, the squared distance between two states in the concatenated features is the sum of their squared distances in the features of the individual networks, so a kernel of the Euclidean distance on the concatenation is a function of the members' mean squared distance; for a Gaussian kernel it is the product of the members' kernels at a common bandwidth. If the members' squared distances differ from a common metric by fluctuations that are independent across initializations, the mean divides the variance of those fluctuations by the number of members, and the kernel is fitted on a steadier metric than any single network provides.

8 Residual correction and error alignment

Fix the training design, neural mean and selected kernel hyperparameters. The following identities hold without sampling assumptions. We first work with one output coordinate; the vector-valued statements follow coordinatewise.

8.1 A frozen regression operator

Fix a design $X_n = (x_1, \dots, x_n)$, a positive-semidefinite kernel with Gram matrix K, and $\lambda > 0$. Define

$$(Sg)(x) = k(x, X_n)(K + \lambda I)^{-1} g(X_n), \qquad R = I - S. \tag{8}$$

Here λ is the diagonal regularizer in the linear system. The experiments use $\lambda = n\gamma$, where γ is the reported nugget; the distinction matters when the training size changes. Although a selected kernel can depend on the targets, S is linear in the response only *after that selection is frozen.*

Proposition 2 (Frozen residual correction)**.** *Let f be the target and h any fitted mean, and set $\widehat{f}_c = h + S(f - h)$. Then:*

$$f - \widehat{f}_c = R(f - h), \tag{9}$$

$$(Rg)(X_n) = \lambda(K + \lambda I)^{-1} g(X_n). \tag{10}$$

If $Ku_j = \mu_j u_j$, the design residual in direction u_j is multiplied by $\lambda/(\mu_j + \lambda) \in (0, 1]$. The factor is one when $\mu_j = 0$ and strictly below one when $\mu_j > 0$; this is shrinkage, not a hard spectral projection.

Suppose, in addition, that $\mathcal{S}$ is a normed vector space containing f and h and that the following bound holds uniformly *on that space for the realized operator:*

$$\|Rg\|_{L^2(P)} \le C_n \|g\|_{\mathcal{S}} \quad (g \in \mathcal{S}).$$

Then

$$\|f - \widehat{f}_c\|_{L^2(P)} \le C_n \|f - h\|_{\mathcal{S}}. \tag{11}$$

Consequently, $\|f - h\|_{\mathcal{S}} < \|f\|_{\mathcal{S}}$ makes this displayed upper bound smaller than $C_n \|f\|_{\mathcal{S}}$ when $C_n > 0$. It is neither a necessary nor a sufficient condition for the corrected predictor's actual error to be smaller than that of Sf.

Proof in Section S9.

Why the bounds do not order the errors. On a two-point design take $K = \mathrm{diag}(1, 99)$ and $\lambda = 1$, so $R = \mathrm{diag}(1/2, 1/100)$ and $C_n = 1/2$ is its Euclidean operator norm. With $f = (1, 0)$ and $f - h = (0, 2)$, the residual norm is twice the target norm, yet $\|R(f - h)\| = 0.02 < 0.50 = \|Rf\|$. Conversely, with $f = (0, 1)$ and $f - h = (1/2, 0)$, the residual norm is smaller, but $\|R(f - h)\| = 0.25 > 0.01 = \|Rf\|$. These examples also show why source-norm accuracy alone does not order corrected prediction errors.

8.2 An exact criterion for improvement

The relevant comparison can be made exactly in any Hilbert error metric. This includes a finite weighted squared error and $L^2(P)$, but not directly the mean of per-sample relative norms or the nonlinear radiance and reflectance metrics in Table 1.

Proposition 3 (Error alignment)**.** *For the same frozen operator S, put $a = Rf$ and $b = Rh$. Then*

$$\|f - \widehat{f}_c\|^2 - \|f - Sf\|^2 = \|b\|^2 - 2\langle a, b\rangle. \tag{12}$$

Thus full correction improves on Sf exactly when $2\langle a, b\rangle > \|b\|^2$. More generally, the interpolation $p_t = Sf + tRh$, $0 \le t \le 1$, has minimum squared error at

$$t_* = \min\{1, \max\{0, \langle a, b\rangle / \|b\|^2\}\} \quad \text{if } b \neq 0; \tag{13}$$

if $b = 0$, every t is equivalent. If the direct fit uses S_0 and the correction uses S_1, (12) *instead holds with $a = (I - S_0)f$ and $b = (S_1 - S_0)f + (I - S_1)h$.*

Proof in Section S9. The criterion depends on the unknown target error. Replacing its inner products by validation errors gives a validation-optimal interpolation, not a guarantee on new data. The reported experiments use separately tuned direct and residual kernels, so their comparison must use the final, two-operator form with their actual selected operators. Neither the criterion nor the training-design contraction establishes that correction will beat both parents at every sample size.

8.3 What the principal-component implementation computes

The pipeline trains a network on all standardized bands but projects its predictions into the retained principal-component basis before correction. Thus its corrected predictor is based on the projected network mean. To state the implementation precisely, let f and h now denote the target and network output after the common fitted standardization and PCA centering. Let Π be the orthogonal projection onto the retained spectral coordinates. In these centered coordinates, the implemented prediction is

$$\widehat{f}_{c,\Pi} = \Pi h + S\Pi(f - h).$$

The output mean and diagonal band scaling are restored afterwards. Since the same scalar kernel operator acts on each retained output coordinate,

$$f - \widehat{f}_{c,\Pi} = (I - \Pi)f + (I - S)\Pi(f - h). \tag{14}$$

The terms on the right are orthogonal in the standardized spectral Euclidean norm at each query, hence their squared norms add there. They need not be orthogonal in physical units after unequal band rescaling. This decomposition separates truncation error from residual regression and explains why output rank is a genuine part of the model specification. It does not by itself prove the observed optimum near rank 32. Comparisons with the unprojected network include the effect of projection as well as residual fitting.

Conditional rate bound. Kernel learning-rate results impose source, spectral, sampling and regularization conditions [47, 48, 49]. If a uniform residual bound holds with $C_n = O(n^{-\alpha})$ and $\|f - h_n\|_{\mathcal{S}} = O(n^{-\beta})$, then (11) gives $\|f - \widehat{f}_c\|_{L^2(P)} = O(n^{-(\alpha+\beta)})$. The source-norm bound and the uniformity required for a data-dependent residual are additional hypotheses. They do not follow from empirical network loss. The experiments measure finite-range error slopes, not these source norms or an asymptotic rate.

9 Retrieval-error transfer and the stacking objective

9.1 Transfer of component error into retrieval error

Section 3.1 controls the inverse where the denominator stays away from zero. The theorem below describes a whole evaluation distribution, with entries of vanishing transmission included, with the constant one, and shows that the resulting loss of rate is not an artefact of the argument. Let P be the evaluation distribution over entries (a state, a band, and the surface reflectance when it is varied), in fixed physical units; expectations are conditional on the fitted emulator. Assume

$$0 \le \rho \le R \le 1, \qquad t > 0, \qquad 0 \le s \le S, \qquad RS < 1. \tag{15}$$

The emulator predicts $(\widehat{a}, \widehat{t}, \widehat{s})$. The *constrained retrieval* replaces $\widehat{t}$ by $\max\{\widehat{t}, 0\}$ and $\widehat{s}$ by its projection onto $[0, S]$, neither of which increases $|e_t|$ or $|e_s|$, and sets

$$\bar{\rho} = \Pi_{[0,R]}\Big(\frac{L - \widehat{a}}{\widehat{t} + \widehat{s}\,(L - \widehat{a})}\Big) \tag{16}$$

when the denominator is positive, and otherwise returns a prescribed value in $[0, R]$ and records a failure. It is an estimator in its own right and is reported beside the unconstrained inverse and its failure counts, never in place of them. The *retrieval-weighted component error*, computed after the projections, is

$$e_R = |e_a| + R\,|e_t| + R^2\,t\,|e_s|, \qquad \varepsilon_R^2 = \mathbb{E}_P[e_R^2], \tag{17}$$

which is at most $e = |e_a| + |e_t| + t|e_s|$ since $R \le 1$; ε_R is not the mean relative radiance error of the tables. Put $F_t(\tau) = \mathbb{P}(t \le \tau)$.

Theorem 4 (error transfer with constant one)**.** *Under* (15)*, for the constrained retrieval* (16)*:*

1. $|\bar{\rho} - \rho| \le \min\{R,\ e_R/t\} \le \min\{R,\ e/t\}$ *at every entry, and no universal constant below one is possible.*

2. *For every* $\tau > 0$*,*

$$\mathbb{E}_P|\bar{\rho} - \rho|^2 \le \mathbb{E}_P \min\Big\{R^2, \frac{e_R^2}{t^2}\Big\} \le R^2 F_t(\tau) + \mathbb{E}_P\Big[\frac{e_R^2}{t^2}\mathbf{1}\{t > \tau\}\Big] \le R^2 F_t(\tau) + \frac{\varepsilon_R^2}{\tau^2}.$$

3. *If* $F_t(u) \le A u^\beta$ *for* $0 < u \le u_0$ *with some* $\beta > 0$*, then for* $\varepsilon_R \le u_0^{(\beta+2)/2}$*,* $\mathbb{E}_P|\bar{\rho} - \rho|^2 \le (R^2 A + 1)\,\varepsilon_R^{2\beta/(\beta+2)}$*; a component-error bound* $\varepsilon_{R,n} = O(n^{-r})$ *yields* $\|\bar{\rho}_n - \rho\|_{L^2(P)} = O(n^{-r\beta/(\beta+2)})$*.*

4. *If* $t \ge t_0 > 0$ P*-almost surely, then* $\|\bar{\rho} - \rho\|_{L^2(P)} \le \varepsilon_R / t_0$*.*

Proof in Section S9. The middle expression in (ii) keeps the association between the errors and the transmission and is the one the tables evaluate; the last needs no power law. Both are computable on a finite test table; (iii) is a population statement whose tail hypothesis a finite table cannot establish down to $t = 0$. The exponent in (iii) cannot be improved from its hypotheses. Item (i) holds entry by entry, so it passes to quantiles: on any set of entries, a single band for instance, every quantile of $|\bar{\rho} - \rho|$ is at most the same quantile of $\min\{R, e_R/t\}$, which depends only on the component errors and the true transmission. In Bayesian inversion the corresponding statements bound the Hellinger distance between the true posterior and one built on a Gaussian-process emulator by moments of the emulator's error [66]; here the inverse is deterministic and the weight of each entry, $1/t$, is explicit.

Proposition 5 (the exponent is attained)**.** *For every $\beta > 0$, every $0 < c < r_0 < R$ and every $\tau \in (0, 1)$ there are an evaluation distribution and a predictor satisfying* (15) *with $S = 0$ and $F_t(u) = u^\beta$ on $(0, 1)$, namely $\rho = r_0$, $a = s = 0$, $\widehat{t} = t$, $\widehat{s} = 0$ and $\widehat{a} = ct\,\mathbf{1}\{t \le \tau\}$, for which $\bar{\rho} = r_0 - c\,\mathbf{1}\{t \le \tau\}$,*

$$\varepsilon_R^2 = \frac{c^2\beta}{\beta + 2}\,\tau^{\beta+2}, \qquad \mathbb{E}_P|\bar{\rho} - \rho|^2 = c^2\tau^\beta,$$
$$\mathbb{E}_P|\bar{\rho} - \rho|^2 = c^{4/(\beta+2)}\Big(\frac{\beta + 2}{\beta}\Big)^{\beta/(\beta+2)}\varepsilon_R^{2\beta/(\beta+2)}.$$

Proof in Section S9. This is sharpness of the transfer inequality for the physical inverse, not a minimax statement over retrieval algorithms. What decides the rate is where the errors sit relative to the transmission.

Theorem 6 (conditional error profile)**.** *Under* (15)*, suppose $\mathbb{E}_P[e_R^2 \mid t] \le \sigma^2 t^{2\nu}$ P-almost surely for some $0 \le \nu \le 1$. Then*

$$\mathbb{E}_P|\bar{\rho} - \rho|^2 \le \mathbb{E}_P \min\{R^2,\ \sigma^2 t^{-2(1-\nu)}\}.$$

Under the tail hypothesis of Theorem 4(iii) and $\nu < 1$, as $\sigma \to 0$,

$$\mathbb{E}_P|\bar{\rho} - \rho|^2 = \begin{cases} O(\sigma^{\beta/(1-\nu)}), & \beta < 2(1-\nu), \\ O(\sigma^2 \log(1/\sigma)), & \beta = 2(1-\nu), \\ O(\sigma^2), & \beta > 2(1-\nu), \end{cases}$$

and at $\nu = 1$, $\mathbb{E}_P|\bar{\rho} - \rho|^2 \le \sigma^2$ with no tail hypothesis. In the model of Proposition 5 the predictor $\widehat{a} = a + \min\{ct, \sigma t^\nu\}$ has $e_R \le \sigma t^\nu$ and $|\bar{\rho} - \rho| = \min\{c, \sigma t^{\nu-1}\}$, and attains each order.

Proof in Section S9. At $\nu = 0$ the theorem recovers the uniform-error regimes under the weaker hypothesis of a bounded conditional second moment; at $\nu = 1$ an emulator whose conditional root-mean-square error decreases proportionally to the transmission loses no rate. The central diagnostic is therefore the profile $t \mapsto \mathbb{E}[e_R^2 \mid t]$ and not only the tail $t \mapsto F_t(t)$: the same overall error concentrated in the opaque entries gives the worst-case loss, and proportional to the transmission gives none.

9.2 The stacking objective

Affine fitting predictions and projected retrieval. Let w denote simplex weights, with one simplex for each component being combined. Write $\widetilde{a}_w$, $\widetilde{t}_w$ and $\widetilde{s}_w$ for the affine combinations

used in fitting, and $\widetilde{e}_a = \widetilde{a}_w - a$, $\widetilde{e}_t = \widetilde{t}_w - t$, $\widetilde{e}_s = \widetilde{s}_w - s$ for their errors. Retrieval uses $\widehat{t}_w = \max\{\widetilde{t}_w, 0\}$ and $\widehat{s}_w = \Pi_{[0,S]}\widetilde{s}_w$, followed by (16). Base predictions may instead be projected before combination, provided the values entering the fitting objective remain affine in w. In either case, projection during retrieval does not increase flux or albedo error under (15).

For a fixed finite threshold $\tau > 0$, the total-flux raw-error surrogate is

$$J_\tau(w) = \mathbb{E}_P\left[\frac{\widetilde{e}_a^2 + R^2\widetilde{e}_t^2 + R^4t^2\widetilde{e}_s^2}{(t \vee \tau)^2}\right]. \tag{18}$$

It is a convex quadratic: the errors are affine in the weights and the entry weights are nonnegative and independent of w. Its empirical version replaces the expectation by the validation average. Theorem 4 and Cauchy–Schwarz imply

$$\mathbb{E}_P|\bar{\rho}_w - \rho|^2 \le R^2F_t(\tau) + 3J_\tau(w). \tag{19}$$

Indeed, projection gives $e_R \le |\widetilde{e}_a| + R|\widetilde{e}_t| + R^2t|\widetilde{e}_s|$; split the expectation at $t = \tau$ and apply the three-term square bound on $\{t > \tau\}$. This inequality holds for every feasible w, not only an optimizer.

A separable surrogate. The stacks of Table 15 combine Y_2 and Y_3 with separate weight vectors. Each base albedo is projected onto $[0, S]$ before fitting, and the base fluxes are not projected. With $\widetilde{e}_j = \widetilde{Y}_j - Y_j$ for $j = 2, 3$, their finite-threshold fitting objective is

$$J'_\tau(w) = \mathbb{E}_P\left[\frac{\widetilde{e}_a^2 + R^2\widetilde{e}_2^2 + R^2\widetilde{e}_3^2 + R^4t^2\widetilde{e}_s^2}{(t \vee \tau)^2}\right]. \tag{20}$$

This is also a convex quadratic. It omits the cross term $2R^2\widetilde{e}_2\widetilde{e}_3$ present in J_τ. The flux floor is applied after fitting, to the combined total flux when the retrieval is scored, and is not part of the fitted loss. Since $t \ge 0$,

$$|\max\{\widetilde{Y}_2 + \widetilde{Y}_3, 0\} - t| \le |\widetilde{e}_2 + \widetilde{e}_3| \le |\widetilde{e}_2| + |\widetilde{e}_3|.$$

The same argument now uses four squared terms and gives

$$\mathbb{E}_P|\bar{\rho}_w - \rho|^2 \le R^2F_t(\tau) + 4J'_\tau(w). \tag{21}$$

For the separable fit the factor is therefore four rather than three. Both statements require the physical hypotheses on the distribution being averaged; full-table statistics that include entries outside those hypotheses are not certificates of either bound.

Putting projection *inside* a squared fitting loss is a different problem and need not preserve convexity. On $0 \le w \le 1$, for example, $g(w) = (\max\{2w - 1, 0\} - 1)^2$ satisfies $g(1/2) = 1 > (g(0) + g(1))/2 = 1/2$. The raw-error quadratic (20) is not of this kind.

Forward-optimal weights. A stack fitted on a forward objective weights every band alike, while the retrieval divides the error at each entry by its transmission. The following example shows how far apart the two optima can be.

Proposition 7 (forward-optimal and retrieval-optimal weights)**.** *Let $0 < \rho < R \le 1$. For a convex combination of heads with weights w, let $F(w)$ be the mean squared radiance error at ρ and $E(w)$ the mean squared error of the constrained retrieval* (16)*, and let w_f minimize F and w_r minimize E.*

1. *Take two equally weighted entries with $a = s = 0$ and transmissions $t_1 = 1$, $t_2 = \delta \in (0, 1]$, and two heads that predict t and s exactly and the path radiance with errors $e^A = (0, c\delta)$ and $e^B = (b, 0)$, where $0 < b, c \le \min\{\rho, R - \rho\}$; w is the weight on A. Then*

$$F(w) = \tfrac{1}{2}\big[(1-w)^2 b^2 + w^2 c^2 \delta^2\big], \qquad E(w) = \tfrac{1}{2}\big[(1-w)^2 b^2 + w^2 c^2\big],$$

$w_f = b^2/(b^2 + c^2\delta^2)$, $w_r = b^2/(b^2 + c^2)$, $F(w_f) \le F(w_r)$, and

$$\frac{E(w_f)}{E(w_r)} = \frac{(b^2 + c^2\delta^4)(b^2 + c^2)}{(b^2 + c^2\delta^2)^2},$$

which tends to $1 + c^2/b^2$ as $\delta \to 0$ and equals $(1 + \delta^2)^2/(4\delta^2)$ at $c = b/\delta$. For every $\tau \le \delta$ the transmission-weighted objective J_τ of (18) equals E.

2. *Suppose instead that $s = 0$ at every entry, that every head predicts t and s exactly, and that the retrieval of every combination lies in $[0, R]$. Let $\mathcal{L}$ be the linear span of the heads' path-radiance errors, and let $\lambda_{\max}$ and $\lambda_{\min}$ be the supremum and the infimum of $\mathbb{E}_P[r^2/t^2]/\mathbb{E}_P[r^2]$ over $r \in \mathcal{L}$ with $\mathbb{E}_P r^2 > 0$. Then*

$$E(w_f) \le \frac{\lambda_{\max}}{\lambda_{\min}}\, E(w_r), \qquad \frac{\lambda_{\max}}{\lambda_{\min}} \le \Big(\frac{t_{\max}}{t_{\min}}\Big)^2 \quad \textit{if} \;\; t_{\min} \le t \le t_{\max} \; P\textit{-almost surely.}$$

Proof in Section S9. The weights that minimize the forward error put almost all their mass on the head whose error sits at the opaque entry, because there it costs little radiance; the retrieval divides that error by δ and pays for it in full. With c/b large the forward-optimal stack is better than either head in radiance and worse than the retrieval-optimal stack by any factor in the inverse, which is the pattern of Table 3 in miniature. Item (ii) says when this can happen. The ratio $\lambda_{\max}/\lambda_{\min}$ measures how unequally the transmission weights the directions of error available to the combination: it equals one when every error in $\mathcal{L}$ has the same transmission profile, and it is large only when the heads put their errors at different transmissions, as the two heads of item (i) do. In that example it equals δ^{-2}, and the ratio $E(w_f)/E(w_r)$ at $c = b/\delta$ comes within a factor of four of it. Without such structure the bound falls back on the squared dynamic range of the transmission. On the physical domain of the EMIT table the transmission itself spans about twenty-four decades, and thirteen above the floor $10^{-12}T_{\text{train}}$ of Section 6.2, so the factor is of order 10^{48} and 10^{26}. The bound (19) for J_τ involves neither quantity.

Numerical weights and threshold convention. The weights are computed by nonnegative least squares with a heavily weighted row penalizing $\sum_h w_h - 1$, and are then normalized to sum to one. This gives feasible simplex weights, though not an exact solution of the equality-constrained quadratic program, and the mean-relative-norm arm runs six iteratively reweighted least-squares sweeps from the squared-loss solution. The weights reported below are therefore approximate minimizers.

A finite threshold is written $\tau = uT_{\text{train}}$, where $T_{\text{train}} > 0$ is a flux scale of the training data, and Table 15 reports u. The unweighted fit is a plain squared-error baseline on the physical components, not the limit of (20) as $\tau \to \infty$. The ladder of thresholds is a sensitivity analysis; a deployed threshold would be fixed in advance by a coverage rule. The stack weights here are fitted on validation rows that also serve the other model decisions, so Proposition S1.2 does not apply to them.

10 Retrieval error and the transmission profile

The transfer theorem makes the ratio of component error to transmission the natural retrieval diagnostic, and the conditional-profile theorem asks for the profile $t \mapsto \mathbb{E}[e_R^2 \mid t]$. This section measures both on finite test sets; a slope fitted to a finite profile is a description of that profile, not an asymptotic rate.

Table 14 uses the three-member width-2000 pipeline on splits 104–107, with float64 predictions of nine heads on 2,331 test states and 285 bands, or 664,335 state–band entries per split. The retrieval uses $\rho = 0.7$, $R = 0.9$ and an albedo bound S taken from the training data.

The physical domain in these splits. S is the largest finite training albedo not exceeding one, and the training blocks hold 227 or 454 entries above one depending on the split. On split 104, 228 of the 664,335 test entries exceed S, and none do on the other three splits; there are 203, 6, 3 and 3 nonpositive transmissions on splits 104, 105, 106 and 107. The pointwise check covers the entries with $s \le S$, and the expectations and ratios in the table are averages over all entries. Section 6 scores the main pipeline on the physical domain itself.

Observed error profiles. The feature kernel has $\varepsilon_R = 0.0514$ and the single network 0.1650. The mean fitted profile slopes 2ν range from 0.16 to 0.28. The raw 95th-percentile error is 0.075 for the single-member feature kernel and 0.060 for concatenated features, against 0.143 for the stack, and the constrained 95th percentiles are 0.076, 0.061 and 0.151. The ratio of the middle bound to the mean squared error lies between about 3.4 and 5.3 across heads.

The middle term of Theorem 4(ii), $\mathbb{E}\min\{R^2, e_R^2/t^2\}$, orders the nine heads by their retrieval tails. Taking the four-split means, its rank correlation with the raw 95th percentile is exactly one (Spearman and Kendall), and with the constrained 95th percentile 0.98 (Spearman), while the unweighted ε_R reaches only 0.78 and 0.81. The stack, for instance, has a smaller ε_R than the kernel on concatenated features (0.060 against 0.063) but a larger middle term (0.036 against 0.030) and a constrained tail more than twice as large. What orders the retrievals is the component error weighted by the transmission, as the theorem says, not the component error itself.

About five percent of the entries have transmission below 3×10^{-4} of the training flux scale, and among positive transmissions the one-percent quantile is of order 10^{-10} of that scale, while the share of squared retrieval-weighted component error on entries below 0.03 of the scale is under a thousandth for every head. Small absolute errors are amplified by weak transmission. The heads still differ in the inverse tail, because the location, sign and dependence of their errors matter, as the exact identities show.

The table's own transmission distribution is extremely heavy near zero: relative to the median positive flux, its 1%, 3%, 5% and 10% quantiles are 1.1×10^{-10}, 3.6×10^{-6}, 2.2×10^{-4} and 1.7×10^{-2}, a log-log slope of $\beta \approx 0.12$–0.13 for F_t over the lowest tenth of the entries. With the measured profile slopes $2\nu \approx 0.16$–0.28, Theorem 6 places the all-band retrieval in its slow regime, $\beta < 2(1-\nu)$, where the mean squared error scales like $\sigma^{\beta/(1-\nu)}$ with exponent 0.13–0.15: halving the component error would shrink it by less than a tenth. On a passband $t \ge t_0 > 0$ fixed in advance the bound of item (iv) is linear in ε_R instead. A slope fitted over a finite range of transmissions does not establish the tail law down to zero or an envelope uniform over training sizes, so this reading describes the table's regime rather than proving a rate. Scaling a fixed error profile down buys little over all bands; moving the error away from the opaque entries, as the feature kernels do, buys much more.

A. Errors and finite-range profiles

head	ε_R	2ν (fit)	raw p95 raw p99	constrained p95 constrained p99
isotropic kernel	0.0886±0.0331	0.16±0.11	0.696±0.000 0.97±0.02	0.266±0.018 0.70±0.00
network	0.1650±0.0177	0.20±0.16	0.670±0.026 0.99±0.14	0.268±0.111 0.70±0.00
network ensemble	0.1085±0.0254	0.18±0.14	0.625±0.013 0.86±0.08	0.239±0.078 0.70±0.01
network + residual	0.0698±0.0373	0.21±0.14	0.489±0.054 0.89±0.01	0.200±0.000 0.70±0.00
ensemble + residual	0.0690±0.0381	0.22±0.14	0.464±0.033 0.88±0.01	0.200±0.000 0.70±0.00
feature kernel	0.0514±0.0454	0.26±0.16	0.075±0.014 0.89±0.05	0.076±0.015 0.70±0.00
concatenated features	0.0634±0.0680	0.28±0.16	0.060±0.004 0.92±0.03	0.061±0.004 0.70±0.00
coordinate selection	0.0721±0.0648	0.25±0.15	0.098±0.031 0.85±0.03	0.099±0.032 0.70±0.00
convex stack	0.0600±0.0661	0.25±0.14	0.143±0.015 1.09±0.05	0.151±0.015 0.70±0.00

B. Failure counts and bound diagnostics

head	failure [%] upper clip [%]	$\mathbb{E}\min\{R^2,$ $e_R^2/t^2\}$	violations on $s \le S$	diagnostic bound / MSE
isotropic kernel	1.09±0.03 3.56±0.07	0.064±0.000	0	3.9±0.2
network	1.31±1.00 2.93±0.43	0.056±0.003	0	3.4±0.7
network ensemble	0.61±0.33 2.56±0.92	0.051±0.002	0	3.7±0.9
network + residual	0.82±0.02 2.51±0.12	0.044±0.001	0	4.0±1.2
ensemble + residual	0.85±0.01 2.43±0.07	0.043±0.001	0	3.9±1.3
feature kernel	0.71±0.02 1.53±0.10	0.031±0.001	0	5.1±2.3
concatenated features	0.72±0.02 1.49±0.02	0.030±0.000	0	5.3±2.4
coordinate selection	0.73±0.02 1.61±0.13	0.033±0.002	0	4.7±2.0
convex stack	0.84±0.05 1.78±0.08	0.036±0.001	0	4.4±2.0

Table 14: Transmission diagnostics for the three-member width-2000 pipeline, splits 104–107. Each cell with a spread is a mean ± sample standard deviation over the four splits. All reflectance errors in this table are in reflectance units, not percentage points. Panel A gives the retrieval-weighted error, the fitted profile slope and the raw and constrained percentiles; panel B gives failure and clipping rates and the bound diagnostics. The violation count tests the pointwise bound on the entries with $s \le S$; on entries with nonpositive transmission or an albedo outside $[0, 1)$ the bound expressions are diagnostics rather than bounds.

11 The stacking objective on data

Table 15 compares stacks fitted under different objectives on five splits: the wide pipeline at seed 101 and the three-member pipeline on splits 104–107. The objectives and the projection sequence are those of Section 9.2.

The mean-relative-norm stack has radiance error 0.0754% and constrained 95th-percentile error 0.141, and the row-relative squared-error stack 0.1130% and 0.180. The finite-threshold J'_τ fits give radiance errors of about 0.22–0.23% for $u \leq 0.1$, 0.1491% at $u = 0.3$ and 0.1175% at $u = 1$, and the unweighted baseline 0.1262%. Weighting by the transmission moves the stack the way Proposition 7 predicts: at $u = 0.003$ the constrained 95th percentile falls from 0.141 to 0.103 while the radiance error rises from 0.075% to 0.216%. It does not reach the kernel on features alone (0.077) or on concatenated features (0.061). The fitting loss and the reported metric need not favor the same combination, and J'_τ is a mean-square criterion while the table reports a quantile; the factor-four transfer bound holds for every feasible weight vector on a distribution satisfying its hypotheses.

12 Discussion

On this table, input scaling and learned features both pay. The input-scaled Matérn kernel reduces radiance error well below the isotropic kernel, the cubic and the width-512 network; residual fitting cuts the network's error by almost a factor of three; and a kernel on learned features matches the input-scaled kernel with one network and beats it with several. The width experiment separates the two roles of a network: the width-2,000 network is the worse emulator and the better feature map. Concatenating the features of an ensemble and solving one kernel on them gives the lowest radiance error of the models fitted to every row and the smallest reflectance tail of the ten-partition comparisons, with one solve in place of a stack.

Forward and inverse rankings still differ. The input-scaled kernel has the smallest median reflectance error of the main comparison, while the feature kernels have the smallest 95th-percentile errors, and both the narrow and the wide stack increase that tail at every split relative to their feature kernels. For the wide model a mean radiance gain of 0.0006 points accompanies an increase from 7.57 to 22.71 points in the mean within-split 95th-percentile reflectance error. Neither a single forward score nor a reflectance median describes this. Section 6 shows that the effect is a property of the emulators: it holds on the physical domain of the retrieval, with coverage and failures reported, after every family is retrained on physically admissible targets only, and on a fresh partition read once under a hypothesis fixed beforehand, where the stack-to-feature-kernel ratios of the tail are 6.60 over all bands and 5.54 on the admissible domain. Retraining on those targets improves the components and not the retrieval: it lowers the albedo error of every family and, at width 512, raises the retrieval tail of every family on average, since the states it removes hold a third of the training entries with the smallest transmissions. On libRadtran, where the transmission stays above 0.14 away from one opaque band, the reversal is much weaker, and what remains of it comes from that band. Both findings fit the account that follows, in which the tail of the retrieval error is set by the component errors at the smallest transmissions.

Theorem 4 explains why such a reversal is possible. On the physical domain the constrained retrieval error is at most e_R/t, capped at R, with constant one, and a distributional form separates the mass of small transmissions from the error relative to transmission on the rest. Proposition 5 shows that the resulting exponent is attained, and Theorem 6 that a conditional

head / fit	radiance [%]	ε_R	constrained p95 constrained p99	raw p95
network	0.8609±0.1051	0.1664±0.0156	0.254±0.101 0.70±0.00	0.677±0.027
network + residual	0.1560±0.0184	0.0654±0.0338	0.200±0.000 0.70±0.00	0.486±0.047
feature kernel	0.0755±0.0114	0.0472±0.0404	0.077±0.013 0.70±0.00	0.076±0.012
isotropic kernel	0.2454±0.0106	0.0886±0.0331	0.266±0.018 0.70±0.00	0.696±0.000
network ensemble	0.5050±0.0592	0.1085±0.0254	0.239±0.078 0.70±0.01	0.625±0.013
ensemble + residual	0.1379±0.0157	0.0690±0.0381	0.200±0.000 0.70±0.00	0.464±0.033
concatenated features	0.0688±0.0128	0.0634±0.0680	0.061±0.004 0.70±0.00	0.060±0.004
stack, mean relative norm	0.0754±0.0116	0.0573±0.0575	0.141±0.033 0.70±0.00	0.133±0.029
stack, row-relative square	0.1130±0.0454	0.0518±0.0414	0.180±0.044 0.70±0.00	0.281±0.122
stack, J'_τ, $u = 0.001$	0.2203±0.1042	0.0633±0.0298	0.135±0.060 0.69±0.01	0.213±0.227
stack, J'_τ, $u = 0.003$	0.2159±0.1100	0.0736±0.0522	0.103±0.055 0.69±0.01	0.182±0.238
stack, J'_τ, $u = 0.01$	0.2303±0.1023	0.0761±0.0508	0.107±0.053 0.59±0.22	0.185±0.236
stack, J'_τ, $u = 0.03$	0.2304±0.1024	0.0761±0.0508	0.107±0.053 0.59±0.22	0.185±0.235
stack, J'_τ, $u = 0.1$	0.2298±0.1013	0.0739±0.0463	0.110±0.052 0.59±0.22	0.188±0.235
stack, J'_τ, $u = 0.3$	0.1491±0.0508	0.0615±0.0468	0.117±0.048 0.69±0.03	0.183±0.196
stack, J'_τ, $u = 1.0$	0.1175±0.0422	0.0553±0.0469	0.156±0.032 0.70±0.00	0.152±0.032
stack, unweighted square	0.1262±0.0406	0.0559±0.0420	0.168±0.039 0.70±0.00	0.194±0.095

Table 15: Stacks under different fitting objectives at $\rho = 0.7$, five splits. The finite-threshold rows use the separable quadratic J'_τ of (20), with $u = \tau/T_{\text{train}}$; the unweighted row is a component squared-error baseline. Radiance is in percent and the reflectance percentiles in reflectance units. Weights are fitted approximately, as described in Section 9.2.

root-mean-square component error proportional to the transmission loses no rate. What decides the retrieval tail is where the component errors sit relative to the transmission, and a stack that lowers the average forward error can move error into the entries with small transmission. The data agree: the transmission-weighted middle term of the bound ranks nine heads exactly as their raw retrieval tails (Section 10), while the unweighted component error does not.

Proposition 7 isolates the mechanism for stacks. Weights that minimize a forward error move mass onto the head whose error sits where the transmission is small, since there it costs little radiance, and the retrieval divides that error by the transmission; the forward-optimal combination can then be worse than the retrieval-optimal one by any factor while winning on radiance. The transmission-weighted quadratic J_τ of Section 9.2 targets the retrieval directly: it bounds the mean squared constrained retrieval error for every feasible weight vector, with factor three (four in its separable form), and in the example of the proposition it is the retrieval error itself. Its weights still come from validation rows, and the bound becomes a guarantee only on an independent selection sample (Proposition S1.2).

For residual correction, the alignment identity and the principal-component decomposition show which parts of the error the correction can remove and which remain. They also show the trade-off behind Table 12, where the corrected network is best at rank 32 and worsens above it while the direct kernel is flat from rank 32 up: a higher rank shrinks the truncation term and hands the kernel more residual coefficients to regress. A learning rate for the corrected predictor would need a uniform regression bound and control of the residual in a source norm; the learning curves measure slopes over a finite range. Proposition S1.3 bounds the movement between two fitted systems; its residual-action form is often much smaller than the spectral bound, though still large relative to the observed movement.

Across the further problems of the supplement the comparison stays metric- and task-dependent. On OCO-2 the coefficient and radiance losses select different representations, on the correction-coefficient benchmark the residual-corrected network improves on the published reference values, and on the ten further configurations the convex stack is at or near the best member of its pool.

Accuracy has a cost. A dense kernel matrix needs order n^2 storage and its factorization order n^3 operations; one $18{,}884 \times 18{,}884$ double-precision matrix occupies 2.85 GB before workspaces. A feature kernel needs a network evaluation and a kernel prediction against the training design, and a stack evaluates every head, so a matched throughput comparison with operational emulators such as sRTMnet [1] remains to be made.

The evidence has limits that bear on its reading. The families were developed on this table and the ten splits overlap, so the split-to-split spread describes one table rather than independent data; the fresh partition guards against the choice of split and of hypothesis, not against the table itself. The table is an adaptively explored lookup table, and an operational evaluation would fix its passband and coverage in advance, separate physical from numerical failures, and report both raw and constrained retrievals, as Section 6 does for the present emulators. Within that scope, neural features and exact kernels complement one another, and a small forward gain can come with a much larger inverse-error tail.

Acknowledgements and funding

The research presented in this paper was supported by the European Research Council (ERC) under the European Union's Horizon Europe research and innovation programme (grant agreement No. 101041711), by the Simons Foundation as part of the Collaboration on the Mathematical and Scientific Foundations of Deep Learning, by Heights Labs, by Convex Nexus Capital, by the Israel Science Foundation (grant number 2258/19), by the Israel Science Foundation (ISF Grant 4101/25), and by Google Cloud through its education credits program (credit 530550103).

This work was done by multiple AI agents, Claude (Anthropic) and Codex (OpenAI), connected to Math Brain [110] and working together inside the Brain framework [112], a shared, persistent memory in which the agents record claims, proofs, failed attempts and methods, retrieve one another's results, coordinate who works on what, and audit each other's work. For more on the details behind the Brain and Math Brain, see the Math Brain paper [111], the Math Brain repository [110] and the Brain framework repository [112].

Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital; both entities supported this work and are named above. These entities and the public funders had no role in the study design, analysis, manuscript preparation, or decision to publish.

AI-assisted contributions

Claude Code (Anthropic) and Codex (OpenAI) implemented the data pipeline and the model comparisons, ran the experiments, produced the figures and tables from the run records, and drafted the manuscript. ChatGPT (OpenAI) developed the residual-error and inversion-stability arguments, checked the identities and wrote the conditioning diagnostics and their tests. The author supplied the data, set the questions, and reviewed the results, the proofs and the text.

Data and code availability

The repository `https://github.com/yspennstate/emit-oco2-kernel-emulators` holds the code, the per-split run records, the scripts that generate every table and figure from those records, and this manuscript. Each record carries the SHA-256 digests of the input arrays, the split sizes, the selected hyperparameters and the test metrics; the records of the training-target experiment also carry the digests of their split indices, and the float64 test predictions of every family in those runs are available on request. The EMIT table is the file `data_EMIT_24k.jld2` (213,761,068 bytes, SHA-256 `344a6d0e6d77b9c152de18505befb10c6e7566bcb5ff9bc00e10d260a048c6de`), supplied by the Jet Propulsion Laboratory. Its root holds the state matrix `X` ($6 \times 23{,}313$), the four components `Y1`–`Y4` ($285 \times 23{,}313$ each), the wavelength grid, the index vectors identifying the lookup table's input and component columns, and per-component affine vectors. The drivers read NumPy exports of `X` and `Y1`–`Y4`, whose digests are recorded in every run. The table is available from the author on request, subject to the permission of the Jet Propulsion Laboratory. The OCO-2 arrays and reference predictions are from the release accompanying [3], and the other configurations use the cited public releases. The libRadtran arrays of Section 6.6 are built from the release of [4] by

code/make_libradtran_arrays.py with the option --cats, and their digests are recorded in results/libradtran/MANIFEST_cats.json.

The paper is part of the Math Brain project, whose public repository is https://github.com/yspennstate/math-brain-public [110].

References

[1] P. G. Brodrick, D. R. Thompson, J. E. Fahlen, M. L. Eastwood, C. M. Sarture, S. R. Lundeen, W. Olson-Duvall, N. Carmon, and R. O. Green. Generalized radiative transfer emulation for imaging spectroscopy reflectance retrievals. *Remote Sensing of Environment*, 261:112476, 2021.

[2] D. R. Thompson, V. Natraj, R. O. Green, M. C. Helmlinger, B.-C. Gao, and M. L. Eastwood. Optimal estimation for imaging spectrometer atmospheric correction. *Remote Sensing of Environment*, 216:355–373, 2018.

[3] O. Lamminpää, J. Susiluoto, J. Hobbs, J. McDuffie, A. Braverman, and H. Owhadi. Forward model emulator for atmospheric radiative transfer using Gaussian processes and cross validation. *Atmospheric Measurement Techniques*, 18:673–694, 2025.

[4] M. A. A. Mazid and N. Rishe. Multi-fidelity emulation of atmospheric correction coefficients with physics-guided Kolmogorov–Arnold networks. *Remote Sensing*, 18(11):1826, 2026.

[5] P. Batlle, M. Darcy, B. Hosseini, and H. Owhadi. Kernel methods are competitive for operator learning. *Journal of Computational Physics*, 496:112549, 2024.

[6] L. Lu, P. Jin, G. Pang, Z. Zhang, and G. E. Karniadakis. Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators. *Nature Machine Intelligence*, 3:218–229, 2021.

[7] Z. Li, N. Kovachki, K. Azizzadenesheli, B. Liu, K. Bhattacharya, A. Stuart, and A. Anandkumar. Fourier neural operator for parametric partial differential equations. In *International Conference on Learning Representations*, 2021.

[8] C. E. Rasmussen and C. K. I. Williams. *Gaussian Processes for Machine Learning*. MIT Press, 2006.

[9] C. Mora, A. Yousefpour, S. Hosseinmardi, H. Owhadi, and R. Bostanabad. Operator learning with Gaussian processes. *Computer Methods in Applied Mechanics and Engineering*, 434:117581, 2025.

[10] A. G. Wilson, Z. Hu, R. Salakhutdinov, and E. P. Xing. Deep kernel learning. In *Proceedings of the 19th International Conference on Artificial Intelligence and Statistics*, pages 370–378, 2016.

[11] G. R. Yoo and H. Owhadi. Deep regularization and direct training of the inner layers of neural networks with kernel flows. *Physica D: Nonlinear Phenomena*, 426:132952, 2021.

[12] Y. Shmalo. Neural means and kernel corrections for operator learning. Preprint, ResearchGate, 2026. Code and per-run summaries at https://github.com/yspennstate/neural-means-kernel-corrections.

[13] R. O. Green et al. The Earth surface mineral dust source investigation: an Earth science imaging spectroscopy mission. In *IEEE Aerospace Conference*, 2020.

[14] E. F. Vermote, D. Tanré, J. L. Deuzé, M. Herman, and J.-J. Morcrette. Second simulation of the satellite signal in the solar spectrum, 6S: an overview. *IEEE Transactions on Geoscience and Remote Sensing*, 35(3):675–686, 1997.

[15] B. Mayer and A. Kylling. Technical note: the libRadtran software package for radiative transfer calculations — description and examples of use. *Atmospheric Chemistry and Physics*, 5:1855–1877, 2005.

[16] C. Emde, R. Buras-Schnell, A. Kylling, B. Mayer, J. Gasteiger, U. Hamann, J. Kylling, B. Richter, C. Pause, T. Dowling, and L. Bugliaro. The libRadtran software package for radiative transfer calculations (version 2.0.1). *Geoscientific Model Development*, 9:1647–1672, 2016.

[17] A. Berk, P. Conforti, R. Kennett, T. Perkins, F. Hawes, and J. van den Bosch. MODTRAN6: a major upgrade of the MODTRAN radiative transfer code. In *Proceedings of SPIE 9088, Algorithms and Technologies for Multispectral, Hyperspectral, and Ultraspectral Imagery XX*, 2014.

[18] C. D. Rodgers. *Inverse Methods for Atmospheric Sounding: Theory and Practice*. World Scientific, 2000.

[19] A. Eldering, C. W. O'Dell, P. O. Wennberg, D. Crisp, M. R. Gunson, et al. The Orbiting Carbon Observatory-2: first 18 months of science data products. *Atmospheric Measurement Techniques*, 10:549–563, 2017.

[20] D. Crisp, H. R. Pollock, R. Rosenberg, L. Chapsky, R. A. M. Lee, et al. The on-orbit performance of the Orbiting Carbon Observatory-2 (OCO-2) instrument and its radiometrically calibrated products. *Atmospheric Measurement Techniques*, 10:59–81, 2017.

[21] B. J. Connor, H. Boesch, G. Toon, B. Sen, C. Miller, and D. Crisp. Orbiting Carbon Observatory: inverse method and prospective error analysis. *Journal of Geophysical Research: Atmospheres*, 113:D05305, 2008.

[22] C. W. O'Dell, B. Connor, H. Bösch, D. O'Brien, C. Frankenberg, et al. The ACOS CO_2 retrieval algorithm — part 1: description and validation against synthetic observations. *Atmospheric Measurement Techniques*, 5:99–121, 2012.

[23] F. Chevallier, F. Chéruy, N. A. Scott, and A. Chédin. A neural network approach for a fast and accurate computation of a longwave radiative budget. *Journal of Applied Meteorology*, 37:1385–1397, 1998.

[24] V. M. Krasnopolsky, M. S. Fox-Rabinovitz, and D. V. Chalikov. New approach to calculation of atmospheric model physics: accurate and fast neural network emulation of longwave radiation in a climate model. *Monthly Weather Review*, 133:1370–1383, 2005.

[25] M. A. Veerman, R. Pincus, R. Stoffer, C. M. van Leeuwen, D. Podareanu, and C. C. van Heerwaarden. Predicting atmospheric optical properties for radiative transfer computations using neural networks. *Philosophical Transactions of the Royal Society A*, 379:20200095, 2021.

[26] P. Ukkonen. Exploring pathways to more accurate machine learning emulation of atmospheric radiative transfer. *Journal of Advances in Modeling Earth Systems*, 14:e2021MS002875, 2022.

[27] S. Rasp, M. S. Pritchard, and P. Gentine. Deep learning to represent subgrid processes in climate models. *Proceedings of the National Academy of Sciences*, 115(39):9684–9689, 2018.

[28] N. D. Brenowitz and C. S. Bretherton. Prognostic validation of a neural network unified physics parameterization. *Geophysical Research Letters*, 45:6289–6298, 2018.

[29] J. P. Rivera, J. Verrelst, J. Gómez-Dans, J. Muñoz-Marí, J. Moreno, and G. Camps-Valls. An emulator toolbox to approximate radiative transfer models with statistical learning. *Remote Sensing*, 7(7):9347–9370, 2015.

[30] J. Verrelst, N. Sabater, J. P. Rivera, J. Muñoz-Marí, J. Vicent, G. Camps-Valls, and J. Moreno. Emulation of leaf, canopy and atmosphere radiative transfer models for fast global sensitivity analysis. *Remote Sensing*, 8(8):673, 2016.

[31] J. Vicent Servera, J. P. Rivera-Caicedo, J. Verrelst, J. Muñoz-Marí, N. Sabater, B. Berthelot, G. Camps-Valls, and J. Moreno. Systematic assessment of MODTRAN emulators for atmospheric correction. *IEEE Transactions on Geoscience and Remote Sensing*, 60:1–17, 2022.

[32] J. L. Gómez-Dans, P. E. Lewis, and M. Disney. Efficient emulation of radiative transfer codes using Gaussian processes and application to land surface parameter inferences. *Remote Sensing*, 8(2):119, 2016.

[33] G. Camps-Valls, J. Verrelst, J. Muñoz-Marí, V. Laparra, F. Mateo-Jiménez, and J. Gómez-Dans. A survey on Gaussian processes for earth-observation data analysis. *IEEE Geoscience and Remote Sensing Magazine*, 4(2):58–78, 2016.

[34] J. Sacks, W. J. Welch, T. J. Mitchell, and H. P. Wynn. Design and analysis of computer experiments. *Statistical Science*, 4(4):409–423, 1989.

[35] M. C. Kennedy and A. O'Hagan. Predicting the output from a complex computer code when fast approximations are available. *Biometrika*, 87(1):1–13, 2000.

[36] M. C. Kennedy and A. O'Hagan. Bayesian calibration of computer models. *Journal of the Royal Statistical Society: Series B*, 63(3):425–464, 2001.

[37] P. Perdikaris, M. Raissi, A. Damianou, N. D. Lawrence, and G. E. Karniadakis. Nonlinear information fusion algorithms for data-efficient multi-fidelity modelling. *Proceedings of the Royal Society A*, 473:20160751, 2017.

[38] B. Peherstorfer, K. Willcox, and M. Gunzburger. Survey of multifidelity methods in uncertainty propagation, inference, and optimization. *SIAM Review*, 60(3):550–591, 2018.

[39] K. Cawse-Nicholson, P. A. Townsend, D. Schimel, A. M. Assiri, P. L. Blake, et al. NASA surface biology and geology designated observable: a perspective on surface imaging algorithms. *Remote Sensing of Environment*, 257:112349, 2021.

[40] N. Aronszajn. Theory of reproducing kernels. *Transactions of the American Mathematical Society*, 68:337–404, 1950.

[41] G. Kimeldorf and G. Wahba. Some results on Tchebycheffian spline functions. *Journal of Mathematical Analysis and Applications*, 33(1):82–95, 1971.

[42] G. Wahba. *Spline Models for Observational Data*. SIAM, 1990.

[43] B. Schölkopf, R. Herbrich, and A. J. Smola. A generalized representer theorem. In *Proceedings of the 14th Annual Conference on Computational Learning Theory*, pages 416–426, 2001.

[44] A. E. Hoerl and R. W. Kennard. Ridge regression: biased estimation for nonorthogonal problems. *Technometrics*, 12(1):55–67, 1970.

[45] H. W. Engl, M. Hanke, and A. Neubauer. *Regularization of Inverse Problems*. Kluwer, 1996.

[46] M. L. Stein. *Interpolation of Spatial Data: Some Theory for Kriging*. Springer, 1999.

[47] A. Caponnetto and E. De Vito. Optimal rates for the regularized least-squares algorithm. *Foundations of Computational Mathematics*, 7(3):331–368, 2007.

[48] D. J. Sutherland. Fixing an error in Caponnetto and de Vito (2007). arXiv:1702.02982v2, 2021.

[49] S. Fischer and I. Steinwart. Sobolev norm learning rates for regularized least-squares algorithms. *Journal of Machine Learning Research*, 21(205):1–38, 2020.

[50] S. Smale and D.-X. Zhou. Learning theory estimates via integral operators and their approximations. *Constructive Approximation*, 26:153–172, 2007.

[51] B. Schölkopf and A. J. Smola. *Learning with Kernels*. MIT Press, 2002.

[52] I. Steinwart and A. Christmann. *Support Vector Machines*. Springer, 2008.

[53] T. Hastie, R. Tibshirani, and J. Friedman. *The Elements of Statistical Learning*. Springer, second edition, 2009.

[54] C. A. Micchelli and M. Pontil. On learning vector-valued functions. *Neural Computation*, 17(1):177–204, 2005.

[55] M. A. Álvarez, L. Rosasco, and N. D. Lawrence. Kernels for vector-valued functions: a review. *Foundations and Trends in Machine Learning*, 4(3):195–266, 2012.

[56] H. Owhadi. Bayesian numerical homogenization. *Multiscale Modeling & Simulation*, 13(3):812–828, 2015.

[57] H. Owhadi and C. Scovel. *Operator-Adapted Wavelets, Fast Solvers, and Numerical Homogenization*. Cambridge University Press, 2019.

[58] Y. Chen, B. Hosseini, H. Owhadi, and A. M. Stuart. Solving and learning nonlinear PDEs with Gaussian processes. *Journal of Computational Physics*, 447:110668, 2021.

[59] H. Owhadi and G. R. Yoo. Kernel flows: from learning kernels from data into the abyss. *Journal of Computational Physics*, 389:22–47, 2019.

[60] B. Hamzi and H. Owhadi. Learning dynamical systems from data: a simple cross-validation perspective, part I: parametric kernel flows. *Physica D: Nonlinear Phenomena*, 421:132817, 2021.

[61] C. K. I. Williams and M. Seeger. Using the Nyström method to speed up kernel machines. In *Advances in Neural Information Processing Systems 13*, pages 682–688, 2001.

[62] P. Drineas and M. W. Mahoney. On the Nyström method for approximating a Gram matrix for improved kernel-based learning. *Journal of Machine Learning Research*, 6:2153–2175, 2005.

[63] F. Bach. Sharp analysis of low-rank kernel matrix approximations. In *Proceedings of the 26th Annual Conference on Learning Theory*, pages 185–209, 2013.

[64] A. Rahimi and B. Recht. Random features for large-scale kernel machines. In *Advances in Neural Information Processing Systems 20*, pages 1177–1184, 2008.

[65] N. H. Nelsen and A. M. Stuart. The random feature model for input-output maps between Banach spaces. *SIAM Journal on Scientific Computing*, 43(5):A3212–A3243, 2021.

[66] A. M. Stuart and A. L. Teckentrup. Posterior consistency for Gaussian process approximations of Bayesian posterior distributions. *Mathematics of Computation*, 87(310):721–753, 2018.

[67] A. Rudi, L. Carratino, and L. Rosasco. FALKON: an optimal large scale kernel method. In *Advances in Neural Information Processing Systems 30*, 2017.

[68] G. Meanti, L. Carratino, L. Rosasco, and A. Rudi. Kernel methods through the roof: handling billions of points efficiently. In *Advances in Neural Information Processing Systems 33*, 2020.

[69] J. Quiñonero-Candela and C. E. Rasmussen. A unifying view of sparse approximate Gaussian process regression. *Journal of Machine Learning Research*, 6:1939–1959, 2005.

[70] M. K. Titsias. Variational learning of inducing variables in sparse Gaussian processes. In *Proceedings of the 12th International Conference on Artificial Intelligence and Statistics*, pages 567–574, 2009.

[71] J. Hensman, N. Fusi, and N. D. Lawrence. Gaussian processes for big data. In *Proceedings of the 29th Conference on Uncertainty in Artificial Intelligence*, pages 282–290, 2013.

[72] A. G. Wilson and H. Nickisch. Kernel interpolation for scalable structured Gaussian processes. In *Proceedings of the 32nd International Conference on Machine Learning*, pages 1775–1784, 2015.

[73] J. R. Gardner, G. Pleiss, D. Bindel, K. Q. Weinberger, and A. G. Wilson. GPyTorch: blackbox matrix-matrix Gaussian process inference with GPU acceleration. In *Advances in Neural Information Processing Systems 31*, 2018.

[74] A. G. Wilson and R. P. Adams. Gaussian process kernels for pattern discovery and extrapolation. In *Proceedings of the 30th International Conference on Machine Learning*, pages 1067–1075, 2013.

[75] D. Duvenaud, H. Nickisch, and C. E. Rasmussen. Additive Gaussian processes. In *Advances in Neural Information Processing Systems 24*, 2011.

[76] D. Duvenaud, J. R. Lloyd, R. Grosse, J. B. Tenenbaum, and Z. Ghahramani. Structure discovery in nonparametric regression through compositional kernel search. In *Proceedings of the 30th International Conference on Machine Learning*, pages 1166–1174, 2013.

[77] G. R. G. Lanckriet, N. Cristianini, P. Bartlett, L. El Ghaoui, and M. I. Jordan. Learning the kernel matrix with semidefinite programming. *Journal of Machine Learning Research*, 5:27–72, 2004.

[78] M. Gönen and E. Alpaydın. Multiple kernel learning algorithms. *Journal of Machine Learning Research*, 12:2211–2268, 2011.

[79] N. Cristianini, J. Shawe-Taylor, A. Elisseeff, and J. Kandola. On kernel-target alignment. In *Advances in Neural Information Processing Systems 14*, pages 367–373, 2002.

[80] C. Cortes, M. Mohri, and A. Rostamizadeh. Algorithms for learning kernels based on centered alignment. *Journal of Machine Learning Research*, 13:795–828, 2012.

[81] J. M. Bates and C. W. J. Granger. The combination of forecasts. *Operational Research Quarterly*, 20(4):451–468, 1969.

[82] D. H. Wolpert. Stacked generalization. *Neural Networks*, 5(2):241–259, 1992.

[83] L. Breiman. Stacked regressions. *Machine Learning*, 24:49–64, 1996.

[84] M. Stone. Cross-validatory choice and assessment of statistical predictions. *Journal of the Royal Statistical Society: Series B*, 36(2):111–147, 1974.

[85] G. H. Golub, M. Heath, and G. Wahba. Generalized cross-validation as a method for choosing a good ridge parameter. *Technometrics*, 21(2):215–223, 1979.

[86] S. Arlot and A. Celisse. A survey of cross-validation procedures for model selection. *Statistics Surveys*, 4:40–79, 2010.

[87] K. Hornik, M. Stinchcombe, and H. White. Multilayer feedforward networks are universal approximators. *Neural Networks*, 2(5):359–366, 1989.

[88] D. P. Kingma and J. Ba. Adam: a method for stochastic optimization. In *International Conference on Learning Representations*, 2015.

[89] I. Loshchilov and F. Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2019.

[90] L. Prechelt. Early stopping — but when? In *Neural Networks: Tricks of the Trade*, volume 1524 of *Lecture Notes in Computer Science*, pages 55–69. Springer, 1998.

[91] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov. Dropout: a simple way to prevent neural networks from overfitting. *Journal of Machine Learning Research*, 15:1929–1958, 2014.

[92] R. M. Neal. *Bayesian Learning for Neural Networks*. Springer, 1996.

[93] J. Lee, Y. Bahri, R. Novak, S. S. Schoenholz, J. Pennington, and J. Sohl-Dickstein. Deep neural networks as Gaussian processes. In *International Conference on Learning Representations*, 2018.

[94] A. G. de G. Matthews, M. Rowland, J. Hron, R. E. Turner, and Z. Ghahramani. Gaussian process behaviour in wide deep neural networks. In *International Conference on Learning Representations*, 2018.

[95] Y. Cho and L. K. Saul. Kernel methods for deep learning. In *Advances in Neural Information Processing Systems 22*, pages 342–350, 2009.

[96] A. Jacot, F. Gabriel, and C. Hongler. Neural tangent kernel: convergence and generalization in neural networks. In *Advances in Neural Information Processing Systems 31*, 2018.

[97] S. Arora, S. S. Du, W. Hu, Z. Li, R. Salakhutdinov, and R. Wang. On exact computation with an infinitely wide neural net. In *Advances in Neural Information Processing Systems 32*, 2019.

[98] M. Belkin, S. Ma, and S. Mandal. To understand deep learning we need to understand kernel learning. In *Proceedings of the 35th International Conference on Machine Learning*, pages 541–549, 2018.

[99] A. Radhakrishnan, D. Beaglehole, P. Pandit, and M. Belkin. Mechanism for feature learning in neural networks and backpropagation-free machine learning models. *Science*, 383(6690):1461–1467, 2024.

[100] A. N. Kolmogorov. On the representation of continuous functions of several variables by superpositions of continuous functions of one variable and addition. *Doklady Akademii Nauk SSSR*, 114:953–956, 1957.

[101] Z. Liu, Y. Wang, S. Vaidya, F. Ruehle, J. Halverson, M. Soljačić, T. Y. Hou, and M. Tegmark. KAN: Kolmogorov–Arnold networks. arXiv:2404.19756, 2024.

[102] N. Kovachki, Z. Li, B. Liu, K. Azizzadenesheli, K. Bhattacharya, A. Stuart, and A. Anandkumar. Neural operator: learning maps between function spaces with applications to PDEs. *Journal of Machine Learning Research*, 24(89):1–97, 2023.

[103] L. Lu, X. Meng, S. Cai, Z. Mao, S. Goswami, Z. Zhang, and G. E. Karniadakis. A comprehensive and fair comparison of two neural operators (with practical extensions) based on FAIR data. *Computer Methods in Applied Mechanics and Engineering*, 393:114778, 2022.

[104] M. Raissi, P. Perdikaris, and G. E. Karniadakis. Physics-informed neural networks: a deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019.

[105] M. V. de Hoop, D. Z. Huang, E. Qian, and A. M. Stuart. The cost-accuracy trade-off in operator learning with neural networks. *Journal of Machine Learning*, 1(3):299–341, 2022.

[106] M. Takamoto, T. Praditia, R. Leiteritz, D. MacKinlay, F. Alesiani, D. Pflüger, and M. Niepert. PDEBench: an extensive benchmark for scientific machine learning. In *Advances in Neural Information Processing Systems 35*, 2022.

[107] R. Ohana, M. McCabe, L. Meyer, R. Morel, F. Agocs, M. Beneitez, M. Berger, B. Burkhart, S. Dalziel, D. Fielding, et al. The Well: a large-scale collection of diverse physics simulations for machine learning. In *Advances in Neural Information Processing Systems 37*, pages 44989–45037, 2024.

[108] S. Yu, W. M. Hannah, L. Peng, J. Lin, M. A. Bhouri, R. Gupta, B. Lütjens, J. C. Will, G. Behrens, et al. ClimSim: a large multi-scale dataset for hybrid physics-ML climate emulation. In *Advances in Neural Information Processing Systems 36*, 2023.

[109] G. H. Golub and C. F. Van Loan. *Matrix Computations*. Johns Hopkins University Press, fourth edition, 2013.

[110] Y. Shmalo, *Math Brain: public snapshot, code and tests*, 2026, `https://github.com/yspennstate/math-brain-public`.

[111] Y. Shmalo, *Math Brain: shared, auditable memory for multi-agent mathematical research*, October 2026, `https://github.com/yspennstate/math-brain-public/tree/main/papers/math-brain-2026-10-24`.

[112] Y. Shmalo, *Brain Framework*, public software repository, 2026, `https://github.com/yspennstate/brain-framework-public`.